

A Review of the Effects of Multiple Stressors on Aquatic Organisms and Analysis of Uncertainty Factors for Use in Risk Assessment

Evelyn H. W. Heugens,^{1,3*} A. Jan Hendriks,³ Tineke Dekker,¹ Nico M. van Straalen,² and Wim Admiraal¹

¹Department of Aquatic Ecology and Ecotoxicology, IBED, Faculty of Science, University of Amsterdam, P.O. Box 94084, 1090 GB Amsterdam, The Netherlands; ²Institute of Ecological Science, Vrije Universiteit, De Boelelaan 1087, 1081 HV Amsterdam, The Netherlands; ³Institute for Inland Water Management and Waste Water Treatment, RIZA, P.O. Box 17, 8200 AA Lelystad, The Netherlands

* Correspondence to E.H.W. Heugens, Department of Aquatic Ecology and Ecotoxicology, IBED, Faculty of Science, University of Amsterdam, P. O. Box 94084, 1090 GB Amsterdam, The Netherlands, heugens@science.uva.nl

ABSTRACT: Risk assessment procedures use toxicity tests in which organisms are subjected to chemicals under otherwise constant and favorable experimental conditions. Because variable and suboptimal environmental conditions are common aspects of natural ecosystems, the hazard of underestimation of risk arises. Therefore, an uncertainty factor is used in the extrapolation of results of standard toxicity tests to field situations. The choice for these uncertainty factors is based on little ecological evidence. This review discusses studies on the toxicity of various chemicals to aquatic organisms, modified by temperature, nutritional state and salinity, excluding papers on changes in bioavailability of compounds. Collected data were analyzed quantitatively to evaluate the validity of toxicity data obtained from standard toxicity tests in the laboratory under field conditions. Generally, organisms living under conditions close to their environmental tolerance limits appeared to be more vulnerable to additional chemical stress. Usually, increasing temperature and decreasing food or nutrient level raised toxicity. The influence of salinity was less clear; metal toxicity increased with decreasing salinity, toxicity of organophosphate insecticides increased with higher salinity, while for other chemicals no clear relationship between toxicity and salinity was observed. The interactions can be explained by several physical and physiological processes, acting on factors such as bioavailability, toxicokinetics, and sensitivity of organisms. Quantitative analysis of data indicated that an uncertainty factor for the laboratory to field extrapolation should be smaller than one for an ecosystem in a temperate region, while a factor greater than one would be appropriate for systems nearby discharge points of cooling water. The factor should be greater than one when varying nutritional state is concerned, but smaller than one with respect to salinity. Dependent on the effect parameter used, the differences in toxicity between laboratory and relevant field situations ranged from a factor of 2.6 to 130 and 1.7 to 15 for the two temperature conditions and 1.2 to 10 for nutritional state. A salinity increase from freshwater to marine water decreased toxicity by a factor of 2.1. However, as less extreme salinity changes are more relevant under field conditions, the change in toxicity is probably much smaller. To obtain uncertainty factors that sufficiently protect natural systems without being overprotective, additional research is required.

KEY WORDS: environmental factors, multiple stress, risk assessment, uncertainty factors.

I. INTRODUCTION

Toxicants enter the aquatic environment in various ways. Risk assessment procedures for compounds have been developed to minimize risks for aquatic ecosystems (EPA, 1984). In these procedures, a major role is assigned to standard toxicity tests in the laboratory in which sensitivity of

organisms to individual substances or field samples is determined. Over the last few years, toxicant levels in Dutch waters have declined (RIZA, 1994a, b). Results of standard toxicity tests with these low concentrations of toxic substances sometimes suggest that present pollution levels have no impact on ecosystems. However, the translation of results of standard toxicity tests to field

situations is difficult because many factors, for example, bioavailability of compounds, adaptation to toxicants, etc., are involved. To reduce the chance of underestimating risk, an uncertainty factor is used to extrapolate these results to field situations. Overestimation of risks is also undesirable, as high costs may be involved to meet quality criteria. As the choice for uncertainty factors is arbitrary and often not based on empirical science, the validity of these factors has been questioned (Chapman et al., 1998). It is unclear whether toxicity in the field is greater or smaller than toxicity found in the laboratory (Van Straalen and Denneman, 1989). Arguments for an uncertainty factor greater or smaller than one (toxicity in the field is higher or lower than in the laboratory, respectively) are given in Table 1.

The purpose of this review is to analyze interactions between chemicals and other stressors acting on aquatic organisms. Although the number of studies using terrestrial organisms increases rapidly, too few papers are available at this time to discuss them here. An understanding of interactions is important for the extrapolation of results of toxicity tests conducted in the laboratory to ecosystems and for the design of site-specific quality criteria for toxicants. Following several authors (Bradshaw and Hardwick, 1989; Calow, 1989; Koehn and Bayne, 1989; Sibly and Calow, 1989), stress is defined as an environmental change that affects community structure and ecological functioning of organisms (i.e., growth and reproduction), leading to reduced fitness. Responses to stress are dependent on the level of organization studied, as a stress response at the cellular or molecular level need not become manifest at the level of individuals or populations due to compensatory and regulation mechanisms (Calow,

1989; Parker et al., 1999). In this article, the term 'multiple stress' is used for the situation where an organism is exposed both to a toxicant and to stressful environmental conditions, such as sub-optimal temperature or food supply.

Toxicants and environmental factors can interact in a variety of ways. The bioavailability of compounds may be influenced by changing environmental conditions. For example, sorption of compounds to particulate organic matter may indirectly influence toxicity. Toxicokinetics may be altered as well, for example, higher uptake or detoxication rates may be induced by high temperature or high food levels. Furthermore, the physiological state of an organism may be affected. For instance, food scarcity may change fat, protein, and carbohydrate concentrations in an organism, which may influence the partition of chemicals over the different compartments. Multiple stressors may change an organism's sensitivity in two ways: exposure to a toxic substance may narrow an organism's tolerance range for environmental factors. On the other hand, these factors may influence the toxic effect concentration of compounds. Papers dealing solely with the first type of interaction (e.g., changed bioavailability by pH, water hardness or humic substances) were excluded because this is not the objective of this review.

From the numerous potential environmental stressors, this article considers the influence of temperature, nutritional state, and salinity on the sensitivity of different groups of aquatic organisms to various classes of chemicals. These environmental factors are chosen because they may highly fluctuate in natural systems, causing stress in organisms. Temperature stress in organisms might occur due to climatic extremes or discharges

TABLE 1
Arguments Used in Establishing a Laboratory-Field Uncertainty Factor

–	In the field, biological availability of chemicals is often lower than in laboratory tests
–	In the field, ecological compensation and regulation mechanisms are operating
–	In the field, adaptation to chemical stress may occur
+	Adaptation often entails costs in ecological performance
+	In the laboratory, organisms are tested under optimal and constant conditions
+	In the field, the organisms are exposed to mixtures of many chemicals

Note: A minus indicates a negative argument for an uncertainty factor greater than one, a plus a positive argument.

Van Straalen and Denneman, 1989.

of heated waste water by power plants (Cairns et al., 1975). Discharges from soda works and potassium and coal mines greatly influence salt concentrations in freshwaters, resulting in salinity stress (Dieperink, 1997). In estuaries, fluctuating temperature and salinity may be natural stressors (Forbes, 1991; McLusky et al., 1986). Variable food or nutrient levels are common in natural ecosystems. Food scarcity may even occur in eutrophicated waters, if populations have grown to their carrying capacity (Buikema et al., 1980 [in Klüttgen et al., 1996]; Kooijman and Metz, 1984).

Three previous reviews have considered the interaction between contaminants and temperature (Cairns et al., 1975), contaminants and salinity (Hall and Anderson, 1995) and the combination of these three factors (McLusky et al., 1986). The present article provides information from new literature references published since these reviews. The sections in which this information is given are divided into an introductory section wherein background information on the specific stressor and our expectations concerning possible interactions are given. This is followed by a section with detailed descriptions of individual studies, succeeded by a summary and conclusions section containing general findings and exceptions to the general patterns observed, as well as conclusions regarding our earlier expectations. In addition, the collected data are used to study quantitative relationships that give an indication of the value of uncertainty factors needed to extrapolate results of standard toxicity tests performed in the laboratory to field situations.

II. METHODS

A. Literature Collection

The studies used in this review assessed the toxicity of a chemical to a certain species at a range of values for one or more environmental factors. Only studies with constant experimental conditions, apart from the factors of interest, were included. The databases Current Contents Search and Biological Abstracts were used to acquire relevant literature. References cited in articles were used as well. As the various toxicants have differ-

ent effects on organisms, chemicals were distinguished into three classes (metals, pesticides, natural toxicants) and a rest group of infrequently studied compounds (other toxicants). In each group the studies were arranged according to compounds and then to trophic groups. When toxic effects increased at rising values of an environmental factor, the factors are said to be positively correlated. A negative correlation indicates decreased toxicity at rising values, and an optimum is designated as the situation where toxicity is lowest or highest at intermediate values of the environmental factor. No correlation means that no systematic effect of the factor on toxicity is observed.

B. Quantitative Relationships

Data were collected from the papers reviewed to study quantitative relationships between toxic effects of contaminants and environmental factors. The data obtained for exposed organisms were scaled to those of the controls. This eliminated the normal effects of temperature, nutritional state, and salinity. By scaling, differences of scale and unit disappeared as well, allowing comparison between experiments. As food quantity and quality varies between studies, food levels in each experiment were scaled to the highest level used in the specific experiment. The relationship between toxicity and temperature is based on the Arrhenius equation and is given by:

$$\log y = \alpha_1 + \beta_T * (1/T) \quad (1)$$

where y is EC50, LC50, or the internal toxicant concentration causing death (lethal body burden), α_1 is the intercept, β_T is a coefficient for temperature, and T is temperature (Kelvin). The relationship between toxicants and nutritional state, salinity, and combinations of environmental factors is given by:

$$\log z = \alpha_2 + \gamma_T * \log T + \gamma_N * \log N + \gamma_S * \log S + \gamma_C * \log C \quad (2)$$

where z is EC50, LC50 or a parameter for growth, reproduction, or survival, α_2 is the intercept, γ_T , γ_N , γ_S , and γ_C are coefficients for temperature, nu-

tritional state, salinity, and toxicant, respectively, T is temperature (Kelvin), N is scaled food or nutrient level (-), S is salinity (‰), and C is toxicant concentration (amount of toxicant/volume). Both linear and multiple regressions were used to fit data to Equations 1 and 2. As data were scarce, studies with limited experimental treatments were still used, although this resulted in less reliable regressions. The coefficients of the environmental factors (β and γ) were arranged in groups according to the effect parameters mostly applied: EC50 (24 h), EC50 (48 h), LC50 (24 h), and LC50(48 h) for temperature and growth, reproduction, survival after chronic exposure, LC50 (48 h), and LC50 (96 h) for nutritional state. Groups were analyzed with a t-test (95% confidence interval). For salinity, too few effect parameters were available to arrange them into groups. Both regressions and statistical analyses were performed by Matlab 5.2 software.

The coefficients of the environmental factors (β and γ) were used to estimate the difference in toxicity under laboratory and relevant field conditions. Standard toxicity tests are usually performed at 20°C, while a much lower temperature can be found in natural ecosystems situated in temperate regions, for example, the Dutch Lake IJsselmeer during the winter season (RIWA, 2000). On the other hand, higher temperatures may occur in systems nearby power plants discharging cooling water (VROM, 1985). Therefore, toxicity at 20°C was compared with toxicity at 3.6 and 30°C. Further, the estimated toxicity at an algal concentration of 3 mg C l⁻¹ was compared with toxicity at concentrations of 0.109 and 0.375 mg C l⁻¹. The first food level is an average food level used for daphnids in standard toxicity tests (OECD, 1998), while the others are the average biomasses of green algae in an oligotrophic and an eutrophic Dutch lake during summer, respectively, Lake Maarsseveen (Hallegraeff, 1976, carbon content calculated from chlorophyll concentration by estimating 2% chlorophyll and 50% carbon on a dry weight basis) and Lake IJsselmeer (Lammens, 1999). For salinity, the change in toxicity was estimated by assuming an extreme salinity increase from 1‰ in the laboratory to 35‰ in the field, which may occur in estuaries (Forbes, 1991).

III. INFLUENCE OF TEMPERATURE

A. Introduction

Most aquatic organisms are ectotherms, which makes temperature an important environmental factor controlling physiological processes. Extreme temperatures may be lethal; the temperature at which death occurs is determined by developmental, genetic, and environmental factors. Within the genetic limits, acclimatization of an organism to a higher or lower temperature can extend the upper or lower boundaries, respectively (Cairns et al., 1975). As temperature influences all kinds of physiological processes, an interaction between temperature and toxicants can be expected when a chemical acts on these processes. For instance, Prosser (1991 [in Rosas and Ramirez, 1993]) proposed that thermal stress acts on the neural membranes, often accompanied by tissue hypoxia. As exposure to toxicants may increase the metabolic oxygen demand, higher temperatures will potentiate the toxic effect. Temperature may also influence toxicokinetics of compounds by affecting metabolic rate or locomotory and feeding activity of an organism, thus affecting uptake, elimination, and detoxication rates (Cairns et al., 1975; Donker et al., 1998; Smit and Van Gestel, 1997). Also, the physiological condition of an organism may be modified by changing temperature, for example, by the induction of proteins protective against cold or heat, which may influence sensitivity to toxicants. At the thermal tolerance limits, exposure to toxicants can enlarge adverse thermal effects (Cairns et al., 1975; Donker et al., 1998). Considering these statements, we expect increased toxicity at elevated temperatures when both stressors act on metabolic processes, such as energy metabolism or respiration. Further, we expect increased internal toxicant concentrations at high temperatures due to elevated toxicant uptake rates, leading to a rise in adverse effects. On the other hand, detoxication and elimination rates may increase as well. It is not clear if this counteracts the thermal effect on toxicant uptake rates.

To examine the influence of temperature on the toxicity of compounds, the critical thermal maximum (CTM) is often used as an indicator of the general physiological state of an organism (Rosas and Ramirez,

1993). The CTM is a measure for the upper limit of thermal tolerance of aquatic animals, which may be modified as a result of toxicant exposure (Alcaraz et al., 1997). The endpoint criterion is the loss of the animal's control over its swimming performance, when it starts to swim vigorously or just before this, when it starts to lose its sense of direction. Comparison of CTM values from different experiments is only possible when a uniform rate of temperature increase and a similar endpoint criterion is chosen. Further, the organisms need to be acclimated to the same temperature (Becker and Wolford, 1980).

B. Detailed Report of Literature

The influence of temperature on toxicity of compounds has been discussed in two reviews. Cairns et al. (1975) summarized the effects of temperature on the toxicity of various chemicals to aquatic organisms, mostly fish, while McLusky et al. (1986) described the effects of temperature and salinity on the sensitivity of marine and estuarine invertebrates to metals. Both studies showed that changing environmental conditions may affect sensitivity of organisms to compounds. In general, toxicity of chemicals increased with rising temperature, and organisms living near to their thermal tolerance limits exhibited greater effect of exposure to toxicants than did organisms living under optimal conditions. Results of studies cited in both reviews are given in *Annex 1*. Studies were only included when mortality was determined, and the number of times that a positive, negative, no correlation, or optimum was found is summed. *Annex 2* gives a summary of the interactive effects of temperature and toxicants found in recent articles reviewed in the present article. Parameters most commonly measured are presented; other parameters and details follow in the subsequent section. Figure 1 presents the frequency of occurrence of positive, negative, no correlations, and optima for effects of toxicants and temperature on mortality in articles discussed in reviews of Cairns et al. (1975) and McLusky et al. (1986) and the present article.

1. Metals

Rosas and Ramirez (1993) found a decreased thermal tolerance in prawn (*Macrobrachium*

rosenbergii) exposed to cadmium and chromium. Compared with control prawns, exposure to 0.05 mg Cd²⁺ l⁻¹ in hard water for 24 h reduced the CTM from 35.8 to 33.8°C. When animals were exposed for 96 h the CTM compared with the control group decreased from 35.4 to 34.6°C. In prawns exposed to 0.7 mg Cr⁶⁺ l⁻¹ in soft water the CTM decreased from 36.1°C for controls to 35.3°C for exposed animals. Exposure time did not significantly influence the decrease in CTM. A reduction in thermal tolerance by cadmium was more prominent in hard than in soft water, while for chromium it was vice versa. A calcium-related uptake of cadmium was assumed; in soft water calcium uptake was reduced, accompanied by a lower cadmium uptake and toxicity. In contrast, the uptake of chromium is not associated with calcium uptake and bioavailability of chromium in hard water is reduced by complexation. It was stated that the observed decrease in thermal tolerance was caused by changes in neurocyte membranes, accompanied by a decrease in oxygen concentration in the tissues.

Carrier and Beitingger (1984) showed that toxicants do not always influence the thermal tolerance of organisms. The CTM of green sunfish (*Lepomis cyanellus*) exposed to cadmium concentrations of 5.17 mg l⁻¹ for 10 days did not significantly differ from that of control fish. This species belongs to the family Centrarchidae, which appears to be resistant to a wide variety of environmental conditions. It was suggested that the lack of response is a result of low cadmium uptake rates, detoxication mechanisms or high cadmium depuration rates.

Jacobson et al. (1997) found increased copper sensitivity of glochidia (free-living larvae) of the freshwater mussel *Actinonaias pectorosa* at elevated temperatures. A rise in temperature from 10 to 25°C resulted in a decrease of LC50 (24 h) from 132 to 42 µg l⁻¹ copper. A positive linear relationship between temperature and LC50 values was found, indicating greater toxicity of the metal at higher temperatures.

Persoone et al. (1989) observed a similar response to increased temperature in *Daphnia magna* exposed to potassium dichromate for 24 h. Dependent on water hardness, exposure to potassium dichromate combined with a temperature increase of 7 to 28°C caused a rise in the amount of immobilized animals by a factor of 3 to 12.

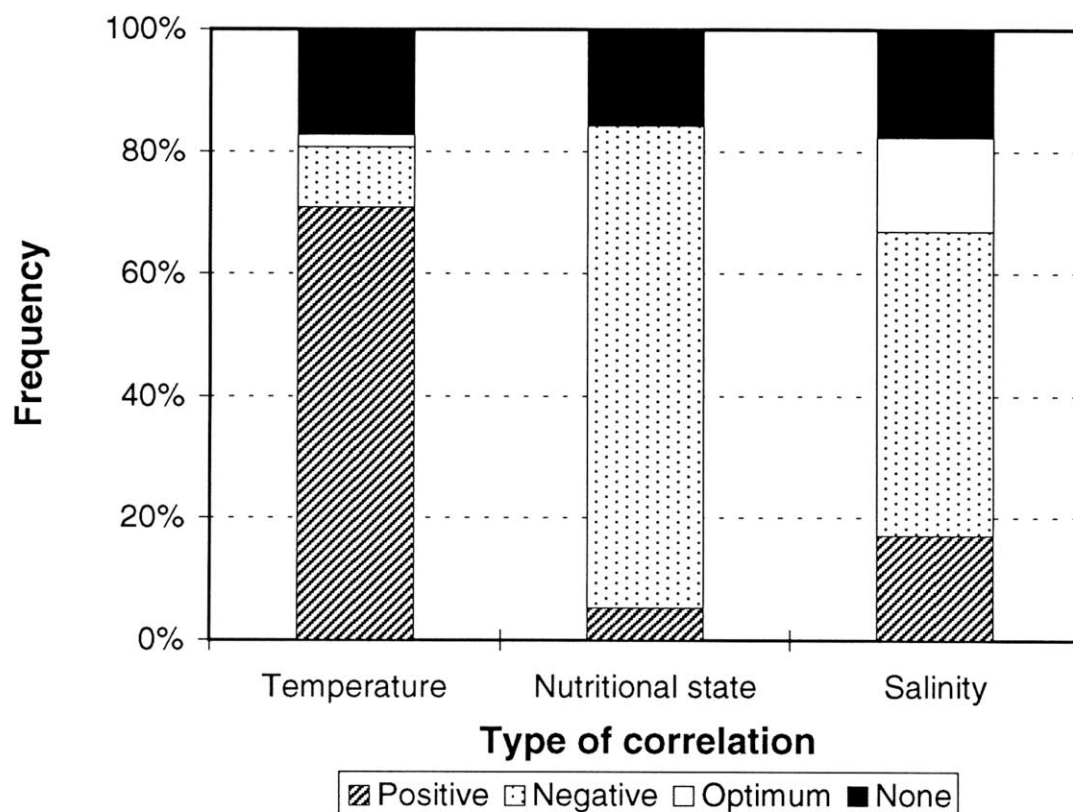


FIGURE 1. Frequency of occurrence of four categories of correlations between toxicity of compounds and environmental factors (temperature, nutritional state, and salinity) with respect to mortality. The four categories are: *positive correlation*: toxicity increases with increasing values of environmental factor, *negative correlation*: toxicity decreases with increasing values of environmental factor, *optimum*: toxicity is lowest or highest at intermediate values of environmental factor, *no correlation* (indicated as 'none'): environmental factor does not influence toxicity. Number of considered experiments was 151 for temperature, 19 for nutritional state and 181 for salinity. Summarized results of Cairns *et al.* (1975), Hall and Anderson (1995), McLusky *et al.* (1986) and the present review.

Similar results were reported by Larrain *et al.* (1998), who exposed the copepod *Tisbe longicornis* to potassium dichromate for 48 h at 13, 16.5, and 20°C and observed 50% mortality at 25.6, 17.8, and 8.6 mg l⁻¹, respectively. The variability between replicate experiments decreased with increasing temperature.

In natural systems, the presence of toxicants and changes in ambient water temperature do not need to coincide. To simulate this, Becker and Wolford (1980) tested toxicant and temperature stress sequentially. The thermal tolerance of juvenile salmonids was tested 24 to 48 h after exposure to sublethal concentrations of nickel by the use of the CTM method. Temperature and time at loss of control over swimming performance and death were determined. Generally, CTM values for exposed fish were lower and more heterogeneous than those from control fish, suggesting

that each individual reacts differently to toxicant exposure. Exposure of coho salmon (*Oncorhynchus kisutch*) to 16 mg l⁻¹ nickel for 14 days significantly altered the thermal response. The temperatures at loss of control and death were 28.4 and 29.1°C for control fish, decreasing to 25.4 and 26.3°C for exposed fish, respectively. The time at loss of control and death also decreased from 134.1 and 141.9 min (control) to 104.0 and 113.4 min (exposed). Exposure of rainbow trout (*Salmo gairdneri*) to nickel also reduced CTM values compared with those of control fish. For example, an exposure to 1.4 mg l⁻¹ nickel for 21 days resulted in a reduction of temperature at loss of control and death of 26.9 and 27.2°C (control) to 23.2 and 24.7°C (exposed), respectively. Time at loss of control and death decreased from 188.7 and 193.2 min (control) to 153.7 and 167.2 min (exposed). A twofold higher

nickel burden in exposed fish compared with untreated fish explained their low resistance to elevated temperatures.

2. Pesticides

Takle et al. (1983) exposed red shiners (*Notropis lutrensis*) to the dipotassium salt of the herbicide endothal (Aquathol-K). Fish were exposed to nominal concentrations of 1 and 30 mg l⁻¹ acid equivalents for 48 h, but this did not significantly alter the CTM.

Song et al. (1997) tested the influence of two temperatures on the toxicity of four insecticides on *D. magna* and *Aedes aegypti*. Both species were more tolerant to tebufenozide at 20 than at 27°C. LC50 (48 h) values for the water fleas and mosquitoes were, respectively, 17.37 and 0.92 mg l⁻¹ at 27°C, while at 20°C less than 50% mortality occurred during the test period. Tebufenozide is a disrupter of the molting process, and its effects are more pronounced at higher temperatures when metabolic rates, and thus growth and development, are enhanced. *D. magna* was more tolerant to aldicarb, a neurotoxin, at 20 than at 27°C. Corresponding LC50 (48 h) values were 0.74 and 0.075 mg l⁻¹. The shapes of the mortality curves were similar. When *A. aegypti* was exposed to aldicarb, higher mortality was found at 20 than at 27°C, although there was no significant difference in LC50 values at the two temperatures (LC50 [48 h] was 0.27 and 0.29 mg l⁻¹, respectively). Temperature did not affect toxicity of dimethoate and imidacloprid in both species. Fisher and Wadleigh (1985) reported that increased temperature significantly lowered the tolerance of midge larvae (*Chironomus riparius*) to the insecticide lindane. LC50 (24 h) values were 13.43, 6.63, and 6.77 mg l⁻¹ at 15, 25, and 35°C, respectively. In an additional uptake experiment animals were exposed to ¹⁴C-lindane for 24 h. Internal concentrations of 0.0342, 0.0713, and 0.0436 mg kg⁻¹ were found at 15, 25, and 35°C, respectively. The low accumulation at 15°C explains the low toxicity at this temperature. At the higher temperatures, however, a similar response was observed, while less lindane was accumulated at 35 than at 25°C. It was assumed that temperature influences both the ability of lindane to penetrate the nervous system and the sensitivity of the target site.

3. Natural Toxicants

Alcaraz et al. (1997) measured the temperature tolerance of white shrimp *Penaeus setiferus* postlarvae when exposed to ammonia, nitrite, and combinations of these chemicals for 72 h. Ammonia did not affect the temperature response, although the neurotoxic action of ammonia could have masked the temperature effect. Nitrite, however, significantly decreased the temperature tolerance. Concentrations of 50 and 100 mg l⁻¹ NO₂⁻-N caused a decrease in CTM of 5.0 and 8.4%, respectively. The thermal tolerance of shrimp exposed to ammonia and nitrite simultaneously was related to the toxicity ratios of ammonia and nitrite concentrations in the mixtures. The CTM of 38.8°C in control animals decreased to 36.6 and 35.8°C in animals exposed to mixtures of NH₃ and NO₂⁻ in toxicity ratios of 1:1 and 1:2, respectively. Supposedly, the reduction in thermal tolerance was caused by the oxidation of haemocyanin by nitrite and a subsequent low capacity to deliver oxygen to tissues. In addition, the increase in water temperature caused an increase in metabolic rate that could enhance ammonia and nitrite accumulation and their toxic effect.

Cyanobacteria may cause adverse effects to daphnids, either by the production of toxic compounds or by the formation of filaments or colonies that disturb the feeding process (Lampert, 1987). Hietala et al. (1997) noted temperature effects on both acute and chronic toxicity of the toxic cyanobacterium *Microcystis aeruginosa* to different clones of *D. pulex*. In the acute test, immobilization was measured at 19 and 24°C. Thermal effects were seen in nine out of ten clones, where EC50 (48 h) values decreased with increasing temperature. The maximal difference in EC50 was 11-fold. Chronic effects at the two temperatures were determined in two clones, by measuring survival, fecundity, and intrinsic rate of natural increase during 20 days. One clone showed a decrease in fecundity when exposed to cyanobacteria and low temperature. The intrinsic rate of natural increase of exposed animals decreased with rising temperature in both clones. As no additional food was given in the acute experiment, the authors suggest that increased toxicant uptake at high temperature was responsible for

the observed effects. This temperature effect was of less importance in the chronic test, where non-toxic green algae were provided as a food source. Here, the effects could be explained by an increased metabolic energy demand at high temperatures. When feeding was inhibited by the cyanobacteria, the ingested amount of green algae could not maintain optimal growth and reproduction, leading to a reduced intrinsic rate of natural increase.

Hanazato (1991a) exposed *D. ambigua* to a chemical (a kairomone) released by *Chaoborus* (Diptera) larvae, whereupon the tolerance to water temperatures ranging from 5 to 30°C was examined. Daphnids developed helmets as a defense mechanism when held in water conditioned by the predator *Chaoborus*. Temperature determined helmet size, which was maximal at 23°C and became smaller with decreasing temperature. At 5 and 10°C, no helmets were formed. No interaction between temperature and kairomone was found when growth or reproduction were considered, but for mortality an interaction was found: at high temperatures (28 and 30°C) survival rates of daphnids exposed to kairomone decreased greatly compared with unexposed animals. For instance, at 30°C all exposed individuals died before the second day, whereas 42% of the control individuals survived until the seventh day. The high mortality in the exposed group occurred during the juvenile stages in which the daphnids developed helmets. Probably molting to the high-helmeted morph increased the risk of mortality at high temperatures.

4. Other Toxicants

Persoone et al. (1989) studied the influence of temperature on the toxicity of sodium laurylsulfate to *D. magna*. The amount of immobilized daphnids increased with temperature. Dependent on water hardness, a temperature rise of 7 to 28°C resulted in a 5- to 12-fold higher amount of immobilized animals.

Phenol decreased the thermal tolerance of central stoneroller minnow (*Camptostoma anomalum*). Chagnon and Hlohowskyj (1989) acclimated fish to 7.5 and 23°C for at least 2 weeks, whereupon they were subjected to phenol.

The CTM was determined after 48 h and at both acclimation temperatures the thermal tolerance was lowered in fish treated with phenol. Exposure to 12 mg phenol l⁻¹ decreased CTM values from 28.8 (control) to 27.1°C and 35.8 (control) to 32.9°C for fish acclimated at 7.5 and 23°C, respectively. As temperature plays a major role in many physiological processes that depend on normal gill function, it is thought that phenol affects the gill tissue, resulting in the observed decrease in thermal tolerance.

Besides physiological factors, physical factors may be influenced by temperature changes as well, which may alter toxicity. This is illustrated by Van Wezel and Jonker (1998) who used the lethal body burden (LBB) as an effect parameter. The LBB is defined as the molar total body concentration of contaminants causing death. In this article, the influence of temperature on the LBB of 1,2,4-trichlorobenzene in the three-spined stickleback (*Gasterosteus aculeatus*) was determined. At 17.0 and 22.5°C, the LBB was significantly lower than at a temperature of 5°C. The differences were 2- to 2.5-fold. This was explained by a temperature-dependent partitioning of the narcotic compound over different compartments in the sticklebacks. The compound mainly accumulates in storage lipids and membrane lipids. The toxic effect is thought to be determined by the concentration of the chemical in the membranes. The partitioning over storage lipid and water and over membrane lipid and water changes differently with temperature, implying that at lower temperatures more 1,2,4-trichlorobenzene has to be accumulated in the storage lipids before the critical concentration in the membrane lipids is reached, resulting in a higher LBB.

C. Summary and Conclusions

The studies reviewed above show that the relationship between toxicants and temperature is mostly investigated with crustaceans and fish as test organisms. In nearly all experiments thermal stress increased the adverse effects of substances, while results of experiments measuring the CTM indicated that exposure to toxicants decreased the ability of organisms to cope with elevated temperatures. Metal toxicity increased at rising temperatures except in one study with fish. This lack

of effect was species specific, as the fish used belongs to a family known to be less sensitive to adverse environmental conditions. No clear relationship was observed between pesticide exposure and temperature, as positive, negative, and no correlations between temperature and toxicity were found. The thermal tolerance of fish was not changed by endothermal in an acute test. As fish are not the target organisms of this herbicide, the tested concentrations could be too low to cause a response within the exposure period. Temperature did not affect toxicity of the pesticides dimethoate and imidacloprid. The neurotoxic action of these chemicals could mask the thermal effect. Natural compounds were more detrimental at high than at low temperature, except for ammonia. Like dimethoate and imidacloprid, ammonia is capable of causing neurotransmission failure, and temperature effects were thought to be masked. The experiments conducted with less frequently studied substances subscribe the mostly reported finding of increased toxicity at elevated temperatures.

Many authors share our hypothesis that temperature potentiates the toxic effects of chemicals acting on metabolism, for example, energy demand and oxygen uptake and distribution. Further, we expected increased internal toxicant concentrations at elevated temperatures due to higher toxicant uptake rates. This idea is often subscribed, but unfortunately in only two studies the internal concentrations were measured. From those papers it is clear that in addition to higher toxicant accumulation due to increased metabolic rate other processes are involved as well, for example accessibility and sensitivity of the target site for chemicals and temperature dependency of the partitioning of narcotic compounds over body compartments. The reviewed literature did not make clear if detoxication and elimination rates increase at higher temperatures. However, the positive correlation between temperature and toxicants mostly observed suggests that these processes are of less importance.

IV. INFLUENCE OF NUTRITIONAL STATE

A. Introduction

Nutritional state has a great impact on life history traits. In relation to toxicity testing, it may

influence the sensitivity of organisms to toxic compounds, as nutrition has a great impact on the physiological condition of organisms. In addition, sorption of toxicants to food particles may either decrease or increase bioavailability, depending on the route of uptake. Furthermore, metabolic rate is increased by higher food levels, which may influence toxicokinetics. Whether toxicity decreases or increases is determined by the overall effect on uptake, elimination and detoxication rates. As it is unclear which of the processes described above has the highest impact on the toxicity of chemicals, we hope to detect a general finding in the literature published.

When crustaceans are used as test organisms, algae are added as a food source. In most articles, food availability is reported as number of cells ml^{-1} . This density does not contain information on the nutritional value of the algae, as was demonstrated by Naylor et al. (1992), who proposed standardization of food levels using carbon delivered per daphnid per day. Therefore, the results of different experiments can only be compared qualitatively.

B. Detailed Report of Literature

The interaction between toxicants and nutritional state on the parameters mostly used are summarized in Annex 3. Figure 1 presents the frequency of occurrence of positive, negative, no correlations, and optima for joint effects of toxicants and nutritional state on mortality in the articles reviewed in the following section.

1. Metals

Studies investigating the interaction between cadmium toxicity and nutritional state, all used crustaceans as test organisms. Klüttgen and Ratte (1994) exposed *D. magna* chronically to cadmium at different food levels and found higher inhibition of juvenile development at lower food levels, whereas body length and brood size were more affected at high food levels. Compared with controls, exposure to $5 \mu\text{g Cd l}^{-1}$ repressed juvenile development by 17% at 0.8×10^6 cells *Chlorella* day^{-1} , while no significant effect occurred at 16×10^6 cells day^{-1} . Body length decreased by 23

and 5% at high and low food levels, respectively. Brood size was inhibited by 69% at high food concentration, but no effect was observed at the low food level. Especially at the high food levels, mortality was lower in treated groups than in controls, which might be due to hormesis, a non-specific positive reaction to low toxicant concentrations (Stebbing, 1982). The intrinsic rate of natural increase was largely independent of algae concentration. Cadmium was thought to interfere with metabolic processes, indirectly affecting food uptake. The effects at high food levels may be explained by higher cadmium accumulation due to increased metabolism. Effects on juvenile development were only detectable when food was scarce, because juveniles are only stressed by food depletion when algae levels become very low.

Other studies with cadmium were conducted by Chandini (1988a, b, 1989), where *D. carinata* and *Echinisca triserialis* appeared to be more sensitive to cadmium when food levels decreased. At algal concentrations of 0.5, 1.5, and 4.5×10^6 cells *Chlorella* ml⁻¹, LC50 (96 h) values calculated were 70, 58, and 340 µg l⁻¹ for *E. triserialis* and 110, 235 and 480 mg l⁻¹ for *D. carinata* (Chandini, 1988a). Chronic toxicity also depended on the nutritional state of the animals because cadmium was more damaging at low food concentration. For *E. triserialis*, estimated EC50 values for life history parameters such as longevity, life expectancy at birth, total fecundity, net reproductive rate, generation time and intrinsic rate of natural increase were in the range of 2 to 13 and 4 to 21 µg Cd l⁻¹ at low and medium food doses, respectively. At high food level no EC50 values could be assessed in the concentration range tested as mortality was too low (Chandini, 1988b). EC50 values for *D. carinata* held at low food concentrations were lower for reproduction parameters (27 to 30 µg Cd l⁻¹) than those for survival (81 to 162 µg Cd l⁻¹). At intermediate food levels EC50 values rose to 51 to 127 µg Cd l⁻¹, and at high food levels values could only be estimated for fecundity and net reproductive rate at 102 and 87 µg Cd l⁻¹, respectively (Chandini, 1989). Observations of the animals indicated that cadmium caused a disturbance of the feeding process. The decline in survival, growth, and reproduction was thought to be related to increased maintenance costs when

daphnids were exposed to cadmium. At high algal densities the total food intake was not affected as much as at low food levels and cadmium effects were mitigated.

Besides the food concentration during experiments, maternal food levels may alter sensitivity of organisms to toxicants as well. Enserink et al. (1990, 1993) conducted acute (48 h) and chronic (21 d) toxicity tests with *D. magna* exposed to cadmium and chromium. After 48 h exposure to cadmium, mortality was three times higher in juveniles descendant from mothers held at high than at poor food conditions, which was related to body size. At low food levels less but large young were born, while mothers at high food density produced lots of small neonates. It was argued that this size-dependent response was either due to the existence of a detoxication mechanism, because costs for detoxication are more easily paid by large individuals with high energy reserves, or by a more favorable surface to body volume ratio for large animals when the toxic action of the chemical depends on its uptake and distribution. No interactions were found in the other treatments, which makes the authors assume that other factors were of more importance in chronic than in acute experiments. The lack of response in daphnids treated with chromium was associated with a different mode of toxic action of this metal compared with that of cadmium.

Hall et al. (1989a,b) used algae as test organisms. Batch cultures of *Chlamydomonas geitleri* and *Chlorella vulgaris* and chemostat cultures of *Chlorella vulgaris* were exposed to copper under phosphorus- or nitrogen-limited conditions. Although the chemostat cultures were more sensitive, in both kinds of cultures copper exposure induced a decline in cell numbers, which was more pronounced in phosphorus- than in nitrogen-limited cultures. As biomass remained constant, it was suggested that copper caused an uncoupling of cell growth and cell division. Final yield was significantly reduced in phosphorus-limited batch cultures. For *Chlorella*, internal copper concentrations were higher at phosphorus than at nitrogen limitation, probably due to increased cell permeability at low phosphorus concentrations. This was not seen in *Chlamydomonas*. Copper is thought to be detoxicated in intracellular polyphosphate granules, the formation of which

depends on the availability of phosphorus. At low phosphorus levels, the formation of these bodies and thus the detoxication of copper is decreased, resulting in greater sensitivity. Growth could even be further reduced when all available phosphorus is used for detoxication. Nott and Langston (1989) also proposed a role for polyphosphate granules in detoxication of chemicals in *Littorina littorea*, but this hypothesis has not yet been tested. Sunda and Huntsman (1998) found interactive effects of manganese and copper or zinc on *Thalassiosira pseudonana*, a coastal diatom. Both copper and zinc inhibited manganese uptake, resulting in a decreased growth rate at high copper or zinc and low manganese availability.

Koivisto et al. (1992) found an interaction between food level and copper toxicity to five cladoceran species: *D. magna*, *D. pulex*, *D. galeata*, *Bosmina longirostris*, and *Chydorus sphaericus*. In all species, effects of copper on survival were amplified by low food concentrations.

Enserink et al. (1995) found interactive effects of lead and food concentration on size and age at first reproduction of *D. magna*. Lowest observed effect concentrations were independent of food levels. Lead reduced body growth under both food conditions, but this delayed reproduction only at the low food level. Results were explained by increased energy costs for detoxication or repair in daphnids exposed to the metal.

2. Pesticides

Barry et al. (1995) noted that sensitivity of *D. carinata* exposed to the insecticides endosulfan and esfenvalerate for 9 days depended on algal concentration. Food level did not influence the toxicity of endosulfan to daphnids with respect to mortality and length at maturity. However, age at maturity increased for daphnids exposed to pesticide and low food levels, while fecundity in the first brood was more affected at high than at low food levels. The more pronounced effects at higher food levels were thought to be due to enhanced metabolism of endosulfan by daphnids and algae, resulting in endosulfan sulfate that is a toxic and more persistent product, and increased toxicant uptake rates. Mortality in daphnids exposed to

esfenvalerate at high food levels was not significantly different from controls, while there was 100% mortality at low food levels. Fecundity in the first brood and length and age at maturity were only analyzed for the highest food levels. No interaction between insecticide and food level was observed. The higher toxicity at lower food concentrations was explained by a change in bioavailability, as the amount of pesticide adsorbed to algae decreased at low algal density.

3. Natural Toxicants

Reinikainen et al. (1998) studied the effect of food on the tolerance of *D. pulex* to toxic cyanobacteria. When daphnids were exposed to *Microcystis aeruginosa* for 7 days, mortality increased more at low than at high levels of additional, nontoxic food. Egg mortality was also higher in daphnids that were starved before exposure to cyanobacteria than in well-fed daphnids (Reinikainen et al., 1995). Gilbert (1996) exposed rotifers, *Synchaeta pectinata* and two clones of *Brachionus calyciflorus*, to different food conditions in the presence of cyanobacteria, *Anabaena flos-aquae*. In *S. pectinata* exposed to *Anabaena*, both lifespan and fecundity were reduced to a greater extent at low food concentration, with fecundity as the most sensitive parameter. In two separate experiments, *Anabaena* caused a decrease in fecundity of 29 to 90% at food levels of, respectively, 2×10^4 and 3×10^3 cells ml^{-1} and of 61 to 95% at 2×10^4 and 5×10^3 cells ml^{-1} of additional food. Only in one of the *Brachionus* clones an interaction between food and *Anabaena* was observed. Lifespan was affected more at low than at high concentrations of additional food. When fecundity was considered, the same clone showed a tendency to be more affected at low food levels, but there was no significant interaction. *Anabaena* contains an alkaloid neurotoxin, anatoxin-a, which inhibits the feeding process of daphnids. Besides this direct effect, filaments of *Anabaena* may physically interfere with filter feeding. The higher sensitivity to cyanotoxin at lower food levels was explained by the increase in relative concentration of these toxic algae as food levels decrease. In addition, well-fed animals are in a better physiological condition, which may lead to a higher

tolerance to cyanobacteria. Walls et al. (1997) found a clear interaction between food concentration and exposure to cyanobacteria. Exposure of *D. pulex* to *Microcystis* for 48 h caused increased mortality at low food levels. No effect on growth was observed. In the same study, daphnids were exposed to a chemical released by *Chaoborus* (Diptera). Growth was not affected, but, compared with controls, age at first reproduction decreased at the low food level. A negative effect of the chemical on the total number of offspring was observed under high food conditions, whereas this number slightly increased at the low food level. It was not clear why adverse effects were observed at high food levels and positive or indifferent effects at low food levels. The authors suggested that the increase in offspring number in the low food and *Chaoborus* treatment was caused by the earlier start of reproduction. In a similar experiment, Hanazato (1991b) observed helmet formation in *D. ambigua* as a defense against predation when exposed to a compound excreted by *Chaoborus*. Helmets were even formed when food levels became very low, but the size decreased, as well as growth, reproduction, and survival rates. These effects were probably induced by energy loss associated with helmet formation, resulting in a reduced tolerance to food deficiency.

4. Other Toxicants

Klüttgen et al. (1996) studied effects of 3,4-dichloroaniline on *D. magna* and *Ceriodaphnia quadrangula* under different food conditions. Treated *D. magna* held at low food density exhibited a delay in first reproduction compared with controls. At high food levels, *C. quadrangula* treated with the same compound started reproduction sooner compared with unexposed animals. Toxicant-induced inhibition of the feeding process was only observed in *D. magna*, suggesting that the more pronounced effect at low food concentration was caused by insufficient food intake. In both species, the number of offspring was reduced to a greater extent at low than at high food levels. However, in *C. quadrangula* slightly promoting effects occurred in the treatment with highest food and lowest toxicant concentration. As was already discussed (Enserink et al., 1990,

1993), the feeding condition of daphnids may influence the susceptibility of their young to chemicals. Baird et al. (1989) reported similar findings for *D. magna* exposed to 3,4-dichloroaniline. Offspring from mothers cultured at an algal concentration of 0.5 mg C l⁻¹ were about twice as vulnerable as those from mothers cultured at 0.05 mg C l⁻¹. LC50 (48 h) values of 104 and 195 µg l⁻¹ were found for neonates descendant from mothers held at high and low food concentrations, respectively.

C. Summary and Conclusions

The papers summarized above reveal that food or nutrient levels influence the outcome of toxicity tests. For daphnids, there are also transgenerational effects, as the feeding condition of the mother affects the susceptibility of her young to toxicants. Daphnids from well-fed mothers were more sensitive than young from poor-fed mothers. In most studies, a negative relationship between nutritional state and toxicity was observed; decreasing food or nutrient levels resulted in increased toxicity of chemicals. Almost all researches used crustaceans as test organisms, but studies with algae and rotifers showed the same trend. Some studies found another interaction between toxicity and nutritional state. Adverse effects on growth and reproduction were observed in daphnids treated with cadmium under high food conditions. This was probably related to a higher metabolic rate caused by the elevated food level, resulting in an increased toxicant uptake rate. The pesticide endosulfan was more toxic at high food levels, what was explained by an enhanced transformation of the pesticide into a more toxic product. It is also suggested that exudates from cyanobacteria have detrimental effects on daphnids, although this effect declined at high levels of nontoxic algae. Further, exposure to kairomone made daphnids reproduce sooner under low food conditions. The underlying mechanism causing this effect is unclear.

In only one study, differences in the bioavailability of the compound at varying food conditions accounted for the observed results. Most authors explained the increased toxicity at low food levels as a decreased ability to pay the en-

ergy costs for detoxication or repair mechanisms. On the other hand, insufficient food uptake caused by the inhibition of the feeding process itself was also suggested. In addition to the indirect relation between nutritional state and detoxication of chemicals, it was assumed that polyphosphate granules in algae were directly associated with detoxication of contaminants. Summarizing, we can conclude that generally the impact of food or nutrient level on an organism's physiological state is the most important factor determining the toxicity of chemicals.

V. INFLUENCE OF SALINITY

A. Introduction

Salinity fluctuations are common in estuaries, but might also occur in freshwater ecosystems as a result of mining activities. Processes for osmoregulation in freshwater differ from those in marine waters. The blood of freshwater organisms is hypertonic relative to their environment. Water flows into the body due to osmotic pressure and there is a passive loss of ions. Plasma osmolarity is determined by Na^+ and Cl^- ions and the regulation of the ionic balance mainly occurs by active uptake of these ions from the ambient water. In contrast, the blood of marine organisms is hypotonic compared with the ambient water. The osmotic pressure causes inflow of ions and loss of water to the environment. To compensate for these processes, the organisms drink seawater and the excess of ions are excreted by the gills (Vonck, 1999).

The pollution of marine environments occurs mainly in coastal areas (Forbes, 1991), where organisms may be exposed to contaminants and simultaneously experience highly fluctuating salinity. This explains why most literature dealing with the toxicant-salinity interaction considers estuarine organisms. This interaction is complex because salinity may act both on physiological processes as well as on the chemical itself; for example, salinity influences the chemical speciation of metals. A rise in salinity increases the degree of metal complexation, decreasing bioavailability (Riedel et al., 1985; Tsuda et al., 1990) and decreasing toxicity because the free

ionic form of metals is thought to be most toxic (Bury et al., 1999; Karen et al., 1999). In addition, the increase in ionic strength at higher salt concentrations can decrease the activity of active toxic species. Increased ion concentration, associated with the rise in salinity, may also interfere biochemically with the toxicant (Riedel, 1985). However, the decline in toxicity at rising salinity is not equal for all species and is probably caused by different modes of toxic action or pathways of toxicant entry (Bury et al., 1999; Karen et al., 1999). For instance, differences in osmoregulation in fresh and saltwater organisms results in species-specific accumulation and excretion of toxicants. In light of these considerations, our expectation is that metal toxicity is enlarged at low salinities. When the concentration of the free metal ion is measured, or when other chemicals are used, we predict lowest toxicity at the isosmotic point of organisms, where the exchange of water and ions with the ambient water is least.

B. Detailed Report of Literature

Hall and Anderson (1995) analyzed the effects of salinity on the sensitivity of aquatic organisms, mainly fish and crustaceans, to various classes of chemicals. In 55, 27, and 18% of the 173 articles, a negative, positive, and no correlation was reported, respectively. Metal toxicity was generally negatively correlated with salinity. Bioavailability and toxicity were found to correlate with the free metal ion concentration, which is higher at low salinity. No clear trend was observed for organic chemicals, except for organophosphate insecticides where a positive correlation between toxicity and salinity was found. They concluded that genetic, life history, and ecological factors should be considered when evaluating the influence of salinity on toxicity of chemicals. McLusky et al. (1986) reviewed the literature on the interaction between salinity and temperature on the acute toxicity of metals to marine and estuarine invertebrates. In general, acute toxicity increased as salinity declined and animals living near to their salinity limits were less tolerant to metal stress. Effects of salinity were explained by competition of metals with calcium and magnesium at uptake sites. Results of both reviews are

shown in Annex 4. Only studies concerned with mortality as effect parameter are included and positive, negative, or no correlations and optima are indicated. Results of the present study are given in Annex 5, details and less common parameters are discussed in the following text. Figure 1 presents the relative number that a positive, negative, no correlation, and optimum occurs for the effects of toxicants and salinity on mortality in papers summarized in Hall and Anderson (1995), McLusky et al. (1986), and this review.

1. Metals

Lin and Dunson (1993) reported enhanced cadmium toxicity in the mangrove-dwelling fish *Rivulus marmoratus* at low salinity. LC50 (96 h) values for fish held in freshwater and water of 14‰ salinity were 2.96 and 21.12 mg total Cd l⁻¹. When the calculated free cadmium concentration was considered instead of total cadmium concentration, the salinity effect disappeared, indicating that changes in free ion concentration were responsible for the differences in mortality. Effect parameters like growth and mortality are most frequently used, but other parameters might be more sensitive. For instance, Vonck (1999) studied the osmoregulation of flounder (*Platichthys flesus*) when exposed to cadmium. Fish were adapted to freshwater, 50‰ seawater, and full-strength seawater and to water fluctuating from freshwater to seawater and back in cycles of 84 h. After adaptation, cadmium was administered intragastrically. Cadmium effects were influenced by the ambient water salinity and the content of the stomach, which is influenced by drinking in seawater fish. At constant salinities, a cadmium concentration of 15 mg Cd kg⁻¹ fish caused a doubling of ion-transporting chloride cell density in freshwater fish from 158 to 320 cells mm⁻², while in 50‰ and full strength seawater this increase was only 40 and 27%, respectively. Serum osmolarity declined only in freshwater fish from 321 to 304 mOsmol l⁻¹. Cadmium exposure combined with fluctuating salinity resulted in a dose-dependent increase of chloride cell density. Cell densities increased with 82 and 160% when fish were treated with 10 mg Cd kg⁻¹ fish during the freshwater and seawater phase of the salinity cycle,

respectively. This increase in chloride cell density coincided with lower serum cadmium levels compared with fish exposed to the same cadmium concentration but experiencing constant salinity, which suggests that the chloride cells are involved in the removal of cadmium from the fish. The response to cadmium was thought to be concealed by the effects caused by salinity fluctuation, because no effect of the metal on serum osmolarity was observed. It was assumed that the stronger response in freshwater compared with seawater was due to an increase in free cadmium concentration, which is the most toxic form. Shazili (1995) also found cadmium sensitivity to increase at lower salinity. Seabass, *Lates calcarifer*, was acclimated to 5, 15, and 30‰ salinity for 2 weeks, whereafter cadmium was added. A concentration of 10 mg Cd l⁻¹ resulted in median lethal times (LT50) of 16.7, 163.3 and 458.3 h at the three salinities, respectively. LC50 (96 h) values were 1.99, 14.2, and 19.0 mg Cd l⁻¹. The author explained the results by an increased fraction of cadmium present in its free anionic and most toxic form at low salinity.

Wildgust and Jones (1998) noted that an estuarine mysid was most tolerant to cadmium at optimal salinity. After acclimation to different salinities for 5 days, *Neomysis integer* was exposed to the free cadmium ion. Mortality was smallest in the isosmotic point of 20‰ salinity, where the LC50 (96) was 34.4 µg Cd²⁺ l⁻¹. At salinities of 12 and 28‰, these values decreased to 5.3 and 3.9 µg Cd²⁺ l⁻¹, respectively. Differences in toxicity could not be completely explained by differences in cadmium speciation. Probably, the increase in osmotic stress outside the isosmotic point declined cadmium tolerance.

Forbes (1991) found a different effect of salinity on cadmium toxicity. Cadmium effects on growth of the estuarine gastropod *Hydrobia ventrosa*, cultured at 23‰ salinity, were most prominent at a salinity of 33‰, while at 13‰ no response of cadmium was observed as snails did not grow. At a salinity of 23‰, the growth of the control group followed a logistic growth model, while at the other treatments this model was no longer useful because of the highly size-dependent growth response to cadmium and salinity. The low salinity was thought to give such a severe stress that an additional cadmium stress was com-

pletely masked. The high toxicity at 33 ‰ salinity was explained by increased osmotic stress, making the snails less able to compensate the higher metabolic demand induced by cadmium.

2. Pesticides

Song and Brown (1998) exposed salt marsh mosquitoes, *Aedes taeniorhynchus*, and brine shrimps (*Artemia* sp.) to four insecticides (aldicarb, dimethoate, imidacloprid, and tebufenozide). To determine salinity effects, the mosquitoes and shrimps were held in isosmotic and hyperosmotic water, which salinity was three and four times the salinity of the isosmotic water, respectively. Mortality in both species was less under isosmotic conditions than under hyperosmotic conditions. LC₅₀ (72 h) and LC₅₀ (48 h) values were estimated for animals held under isosmotic and hyperosmotic conditions, respectively. For *Aedes* tested at isosmotic and hyperosmotic conditions the following LC₅₀ values were calculated: 0.20 and 0.15 mg aldicarb l⁻¹, 0.20 and 0.031 mg dimethoate l⁻¹, 0.021 and 0.013 mg imidacloprid l⁻¹, and 0.35 and 0.15 mg tebufenozide l⁻¹. For *Artemia*, LC₅₀ values were 17.25 and 5.46 mg aldicarb l⁻¹ and 10.14 and 15.73 mg dimethoate l⁻¹ in isosmotic and hyperosmotic water, respectively. For imidacloprid and tebufenozide, the mortality at isosmotic salinity was too low to estimate LC₅₀ values, but at high-salinity LC₅₀ values were 361.23 and 5.53 mg l⁻¹. Although the LC₅₀ of dimethoate for *Artemia* was higher in the hyperosmotic environment, the total number of animals that died was less at isosmotic conditions. Identical slopes of the mortality response curves of *Aedes* exposed to aldicarb and tebufenozide and of *Artemia* exposed to imidacloprid at the two salinities indicated a similar response to these contaminants at both salinities. The authors assumed that increased physiological costs for osmoregulation decreased the overall fitness of the animals, reducing the tolerance to toxicants.

3. Other Toxicants

Palawski et al. (1985) exposed 35-day-old striped bass (*Morone saxatilis*) to a mixture of 18

organic and inorganic chemicals. Fish were held i.a. in freshwater with high water hardness and in saline water. LC₅₀ (96 h) values at 1 and 5‰ salinity were 1.5 and 3 times higher than in freshwater, respectively. A model was used to calculate the theoretical metal speciation of the inorganic fraction and results indicated that factors like water hardness, alkalinity, salinity, and pH played a major role in metal speciation. The authors attribute the variation in mortality to this speciation in the test media, resulting in a different bioavailability of the metals.

Toxicity of narcotic compounds is supposed to depend on the toxicant concentration in the membranes. Van Wezel and Jonker (1998) reported that salinity had no influence on the lethal body burden of 1,2,4-trichlorobenzene in three-spined stickleback (*Gasterosteus aculeatus*), suggesting that salinity does not influence the direct toxicity of narcotic compounds.

C. Summary and Conclusions

Crustaceans and fish are commonly used as test organisms to investigate toxicity-salinity interactions. These interactions were not clear for all types of chemicals. Metal toxicity was often found to increase at low salinities. Severe osmotic stress at low salinities may explain the lack of cadmium effects on gastropods at low salinity, whereas effects were observed at high salinity. No clear relationship was observed for pesticides. An exception were the organophosphate insecticides, where toxicity increased at rising salinity. This could be caused by increased accumulation at high salinity relative to low salinity. Somasundaram and Coats (1991) and Walker (1976) (both in Ordelman et al., 1994) reported a higher degradation half-life of the organophosphate insecticide malathion in seawater (3 to 5 days) compared with freshwater (1 day).

As expected, the general finding was increased metal toxicity at lower salinity. This was associated with increased bioavailability, as competition of the metals with calcium and magnesium at uptake sites decreases and metal speciation declines with lower salinity. Studies using the free metal ion showed no interaction between the chemical and salinity or minimum toxicity at the

isosmotic point. Except for the organophosphate insecticides, no clear interaction between other chemicals and salinity were observed. These responses were generally explained by increased physiological costs for osmoregulation, decreasing the overall fitness of organisms and making them less capable to endure toxic stress. This supports our hypothesis that toxicity is lowest in the isosmotic point of organisms.

VI. INFLUENCE OF COMBINED FACTORS

A. Introduction

As illustrated above, few studies considered the effects of environmental factors on toxicity, but even less studies have investigated the combined action of these factors. Papers described in this section are dealing with the interaction between two environmental factors and their effect on toxicity. Regarding the statements made in previous sections, toxicity is expected to be potentiated at high temperature, poor food conditions, and salinities outside the organism's isosmotic point. Further, we expect effects of low food levels and suboptimal salinities to become worse at high temperatures, as metabolic needs increase. The lower energy intake and higher physiological costs for osmoregulation will result into a decline in the organism's fitness, increasing its sensitivity to toxicants.

B. Detailed Report of Literature

1. Temperature and Nutritional State

Besides food quantity, food quality is also likely to influence sensitivity of organisms to toxicants. However, Stephenson and Watts (1984) found no interaction between nutritional value and toxicity when *D. magna* was exposed to potassium dichromate for 21 days. Although both temperature and food quality had significant effects on mortality and reproduction, neither 24 h nor 48 h EC50 values for immobilization indicated that temperature or food quality influenced the susceptibility of the daphnids to potassium dichromate.

2. Temperature and Salinity

The effects of temperature and salinity on the acute toxicity of metals have been reviewed by McLusky et al. (1986). Most studies were on crustaceans, with some references on molluscs, annelids, and hydrozoans. In general, cadmium and chromium toxicity increased with rising temperatures and decreasing salinity. One study described effects of copper, lead, mercury, zinc, and nickel on crustaceans. Toxicity increased at high temperature, most notably at high salinity. At constant temperature, toxicity increased at decreasing salinity. The common observation of increased toxicity at high temperature and low salinity was not always seen when organisms were exposed to mercury. However, changes from an organism's optimal condition increased its sensitivity as was reported by several authors. Summarizing, sublethal concentrations at optimal environmental conditions became lethal at suboptimal conditions of temperature, salinity, or both.

McLusky and Hagerman (1987) investigated the survival of the mysid crustacean *Praunus flexuosus* exposed to chromium, nickel, and zinc at temperatures and salinities ranging from 5 to 15°C and 4.5 to 27‰, respectively. Survival decreased at rising temperatures and salinities above or below the isosmotic point (20 to 24‰). Considering the combined action of the two environmental factors, all metals caused minimum toxicity at the isosmotic point and 5°C. Maximum toxicity of chromium and nickel was found at 15°C and 4.5‰ salinity, while zinc caused more adverse effects at the highest salinity. Mortality of animals was linked to a disruption of the osmoregulation process, but this disruption could also be indirectly caused by metal poisoning and lethality. Acclimation to the experimental salinities for over 3 weeks did not result in different responses to the metals.

Persoone et al. (1989) also investigated the combined action of temperature and salinity on toxicity. The brine shrimp *Artemia salina* and the rotifer *Brachionus plicatilis*, a marine and a brackish water organism, were exposed to potassium dichromate and sodium laurylsulfate, a detergent, at different combinations of temperature and salinity. Both temperature and salinity and their interaction affected toxicity, except for *B. plicatilis*

exposed to laurylsulfate, where no significant interaction between the two environmental factors was observed. The effect of salinity on toxicity of dichromate to *A. salina* was temperature dependent; at low temperatures a decrease in salinity resulted in higher toxicity, but at high temperatures the same salinity change caused less effect. Toxicity increased at higher temperatures. Lowest and highest LC50 (24 h) values were 8.8 mg l⁻¹ at 30°C and 20‰ salinity and 291.5 mg l⁻¹ at 10°C and 50‰ salinity, while under standard test conditions (25°C and 35‰ salinity) a LC50 (24 h) of 22.2 mg l⁻¹ was estimated. The toxicity of laurylsulfate increased at higher salinities and temperatures. Lowest and highest LC50 (24 h) values were 7.2 mg l⁻¹ at 30°C and 50‰ and 154 mg l⁻¹ at 10°C and 5‰, respectively. The LC50 at standard conditions was 21.5 mg l⁻¹. Dichromate was more toxic to *B. plicatilis* at lower salinity and higher temperature. Lowest and highest LC50 (24 h) values were 130 mg l⁻¹ at 31°C and 5‰ and 690 at 17°C and 45‰, while under standard conditions a value of 347 mg l⁻¹ was found. Effects of laurylsulfate were more pronounced at high salinity and high temperature, but no interactive effect between these factors was found. Lowest LC50 values were estimated at 24°C and 65‰ (10.9 mg l⁻¹) and highest values at 10°C and 5‰ (29.4 mg l⁻¹). In comparison, the LC50 at standard conditions was 15.4 mg l⁻¹. In short, toxicity of both compounds was highest at high temperatures. Toxicity of dichromate was highest at low salinity, whereas laurylsulfate was more toxic at high salinity.

3. Nutritional State and Salinity

Riedel (1984) studied the toxicity of hexavalent chromium on the estuarine diatom *Thalassiosira pseudonana* as a function of both salinity and sulfate. At a constant salinity of 3.20‰, concentrations causing 50% reduction of growth rate decreased from 6.56 to 0.40 µM chromium at 2.90 and 0.15 mM sulfate, respectively. A 50% reduction in fluorescence yield, a measure for photosystem efficiency, was obtained at 4.94 to 0.34 µM chromium, respectively. At 0.32‰ salinity and sulfate concentrations between 0.38 and 0.02 mM, EC50 values for growth and fluorescence yield ranged from 0.75 to 0.04 and

0.73 to 0.02 µM chromium, respectively. Inhibition was a function of the chromium to sulfate molar ratio: inhibition occurred when this ratio exceeded 500:1. Adverse effects of chromium were thought to be caused by inhibition of sulfate uptake. To confirm this assumption, the relationship between chromium toxicity and chromium and sulfate uptake in *T. pseudonana* was investigated (Riedel, 1985). Chromium affected growth, especially at low sulfate concentrations. Sulfate uptake rates declined at rising chromium concentrations, but it was not clear whether slower uptake reduced growth or if slower growth decreased sulfate uptake. The author speculated that chromium is taken up by sulfate transport systems. The sulfate uptake rate and the ratio of chromium to sulfate would then determine the chromium uptake rate. In the cells, the hexavalent chromium is reduced to trivalent chromium by oxidation of organic molecules. Adverse effects caused by chromate may be due to the oxidation of essential molecules, or the production of trivalent chromium, that has a high affinity for proteins. When cell damage by oxidation or the trivalent chromium concentration in the cells reached a certain threshold growth ceased.

C. Summary and Conclusions

In summary, a small number of studies investigated the influence of combinations of environmental factors on toxicity. Most experiments were conducted with crustaceans. The lack of information on this topic hinders the detection of clear relationships. Food quality and temperature did not influence toxicity. The combined action of temperature and salinity seemed to intensify their separate effects on toxicity. High temperatures and low salinity resulted in major toxic effects. No clear interaction could be observed between nutritional state and salinity; it seemed that toxic effects were mainly related to nutrition and not to salinity. Despite the lack of data, it appears that toxic effects are more pronounced at suboptimal than optimal conditions.

VII. QUANTITATIVE RELATIONSHIPS

If available, data were collected from papers reviewed above and fitted to the models described in equation (1) and (2) given in the methods sec-

tion. The calculated coefficients are given in *Annex 6*. In Table 2, the ranges and averages of the coefficients are given for the effect parameters mostly used.

The two different coefficients for temperature, β_T and γ_T , are not comparable, as different models were used to fit the data. Nevertheless, the positive values for β_T and negative values for γ_T , indicate a positive relation between toxicity and temperature, as β_T is inversely and γ_T directly related to temperature. The β_T coefficients were divided into four groups: EC50 (24 h), EC50 (48 h), LC50 (24 h), and LC50 (48 h). A t-test was performed to reveal which groups were significantly different (95% confidence interval). The results are given in Figure 2A. The thermal effect on toxicity appears to increase with exposure time, although this is not significant for the LC50 values.

The negative relation between nutritional state and toxicity becomes manifest in the positive value of the coefficients (γ_N) for parameters as growth, reproduction, survival after chronic exposure, LC50 (48 h), and LC50 (96 h). The variation in coefficient values for reproduction is probably due to the diversity of effect parameters used. Similar to temperature, exposure time determines to which extent nutritional state influences toxicity. This is depicted for the LC50 values in Figure 2B. The combined action of nutritional state and salinity was determined with growth as effect parameter. The calculated γ_N values are higher compared with those for growth in studies where nutrition is the only environmental variable. Differences in test organisms could account for this variation; in the study with nutrition and salinity, algae were used, while in the others zooplankton was used.

For salinity, only LC50 values are available. The coefficients (γ_s) are mostly positive, indicating decreased toxicity at higher salinities. One exception is a study with the free cadmium ion, where the coefficient has a negative value. Experiments dealing with the combined effects of temperature and salinity on toxicity all used LC50 as effect parameter. Both positive and negative values were found for γ_s . The negative average value is mainly determined by the negative values obtained from experiments with laurylsulfate.

The coefficients for single factors give an indication of the validity of toxicity data obtained from standard toxicity tests under field situations. The standard tests are usually conducted at 20°C, while the temperature in natural ecosystems may be much lower or higher, for instance, in waters situated in temperate regions or receiving discharges of cooling water. Considering the average value of the coefficients calculated for temperature, a temperature reduction of 20 to 3.6°C declines toxicity by a factor of 5.2, 130, 2.6, and 22, while a rise of 20 to 30°C increases toxicity by a factor of 2.5, 15, 1.7, and 5.6 for EC50 (24 h), EC50 (48 h), LC50 (24 h), and LC50 (48 h), respectively. In laboratory tests, food or nutrient availability is *ad libitum*, while this is less favorable under field conditions. Therefore, a food level of 3 mg C l⁻¹, which is used in standard toxicity tests for daphnids (OECD, 1998), was compared with toxicity at 0.375 and 0.109 mg C l⁻¹, which are average biomasses of green algae in an eutrophic and an oligotrophic Dutch lake during the summer season, respectively (Hallegraeff, 1976; Lammens, 1999). According to the average value of the calculated coefficients for nutritional state, this means that toxicity in eutrophic and oligotrophic natural systems would be 1.2 to 1.3, 2.1 to 3.3, 2.1 to 3.2, 1.5 to 1.9, and 4.2 to 10 times higher than toxicity determined under standard test conditions, with respect to growth, reproduction, survival after chronic exposure, LC50 (48 h), and LC50 (96 h), respectively. For salinity, the change in toxicity was estimated by assuming an extreme salinity increase from 1‰ in the laboratory to 35‰ in the field. This results in a 2.1 times lower toxicity under field conditions.

VIII. SYNTHESIS AND DISCUSSION

A. Introduction

Although standard toxicity tests are conducted at constant and favorable experimental conditions, results reported in this paper reveal that variable and suboptimal conditions may modify the outcome of toxicity tests. Elevated ambient temperatures were shown to increase toxicity of various classes of chemicals to aquatic organisms. In accordance, exposure to toxicants decreased the

TABLE 2
Ranges and Averages of Coefficients for Temperature (β_T and γ_T), Nutritional State (γ_N) and Salinity (γ_S) (see Equations 1 and 2), Arranged According to Effect Parameters Mostly Used, n = Number of Experiments

Factor	Effect parameter	n	Coefficient range		Coefficient average $\pm$ standard deviation	
Temperature	EC50 (24 h)	8	2.24*10 ³	to 4.71*10 ³	3.55*10 ³	$\pm$ 8.7*10 ²
	EC50 (48 h)	9	2.20*10 ³	to 1.81*10 ⁴	1.04*10 ⁴	$\pm$ 5.4*10 ³
	LC50 (24 h)	2	1.33*10 ³	to 2.75*10 ³	2.04*10 ³	$\pm$ 1.0*10 ³
	LC50 (48 h)	7	4.47*10 ³	to 1.25*10 ⁴	6.66*10 ³	$\pm$ 2.7*10 ³
Nutritional state	Growth	4	-0.0459	to 0.188	0.0701	$\pm$ 0.096
	Reproduction	13	-0.402	to 2.36	0.357	$\pm$ 0.76
	Survival	9	0.136	to 0.676	0.347	$\pm$ 0.18
	LC50 (48 h)	2	0.131	to 0.253	0.192	$\pm$ 0.086
	LC50 (96 h)	2	0.671	to 0.719	0.695	$\pm$ 0.035
Salinity	LC50 (96 h)	12	-0.0339	to 1.31	0.209	$\pm$ 0.36
Nutritional state and salinity	EC50 ¹	2	0.935	to 0.973 (nutrition)	0.954	$\pm$ 0.27 (nutrition)
			0.982	to 1.08 (salinity)	1.03	$\pm$ 0.069 (salinity)
Temperature and salinity	LC50 (24 h)	7	-0.226	to -2.6 (temperature)	-1.08	$\pm$ 1.1 (temperature)
			-0.582	to 0.447 (salinity)	-0.0948	$\pm$ 0.43 (salinity)

¹Exposure time not reported.

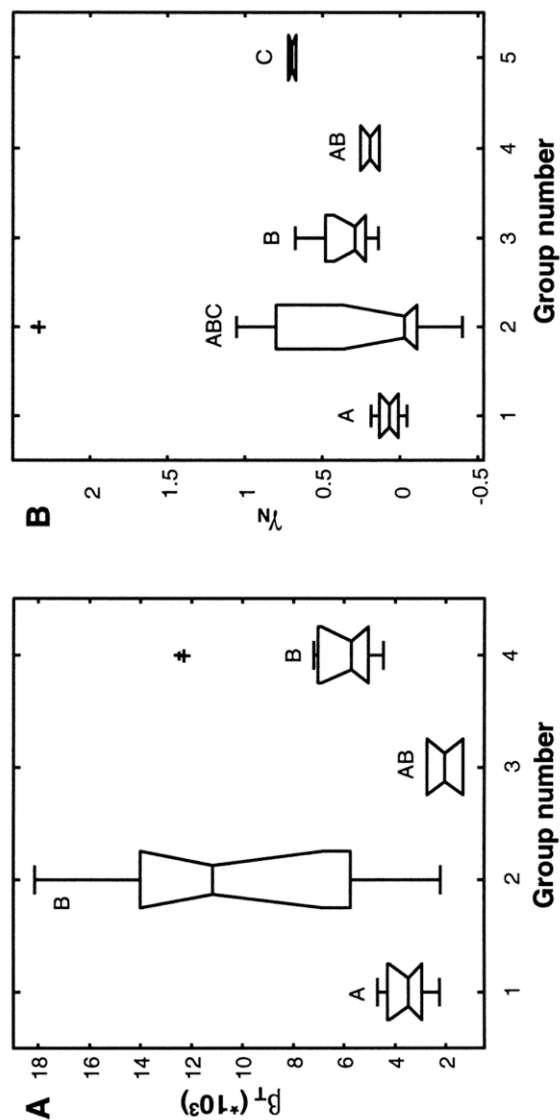


FIGURE 2. Boxplots of temperature coefficients (β_T) for EC50 (24 h) (group 1), EC50 (48 h) (group 2), LC50 (24 h) (group 3), and LC50 (48 h) (group 4) as effect parameter (A) and of coefficients for nutritional state (ζ_N) with respect to growth (group 1), reproduction (group 2), survival after chronic exposure (group 3), LC50(48h) (group 4), and LC50 (96 h) (group 5) as effect parameter (B). The box has horizontal lines at the lower quartile, median, and upper quartile values. The vertical lines (whiskers) show the extent of the rest of the data. The notches give an estimate of the uncertainty about the means for box to box comparison. Values differing more than 1.5 times the interquartile range are outliers, depicted by a plus. Dissimilar letters indicate groups that are significantly different (95% confidence interval).

thermal tolerance. Further, toxicity of contaminants was usually increased under less favorable nutritional conditions. The influence of salinity on toxicity was less clear, although toxic effects of metals were mostly found to increase in less saline waters, whereas toxicity of organophosphate insecticides increased with higher salinity.

Three previous reviews dealt with multiple stressors. Cairns et al. (1975) studied the interaction between contaminants and temperature and concluded that because no generalizations can be made very little evidence is available. Comparing these conclusions with the findings of the present article, it appears that the amount of literature published has increased considerably since. Hall and Anderson (1995) discussed the interactive effects of contaminants and salinity on aquatic organisms and share our conclusion on the influence of salinity on the speciation of metals and the associated effect on toxicity. Further, they concluded that the different toxicity of compounds for marine, brackish, and freshwater organisms is determined by physiological characteristics of the test species. This supports our idea of minimal toxicity at the isosmotic point of organisms, which is species specific. McLusky et al. (1986) reviewed literature dealing with the influence of temperature and salinity on acute toxicity of metals to marine and estuarine invertebrates. They found similar correlations as reported in the present article. Their conclusion that toxicity values obtained under fixed temperature and salinity regimes are unsuitable to evaluate the influence of these environmental conditions on metal toxicity agrees with our results, as well as the view that organisms near to their tolerance limits are less capable to withstand toxic stress. Although the newly reviewed articles did not modify the conclusions made in the previous reviews, we think that this new information provides greater insight into the mechanisms involved in multiple stress interactions. These mechanisms will be discussed in the following sections.

The effects of toxicants on organisms are determined by the bioavailability and toxicokinetics of the compound and the sensitivity of organisms. To understand the interaction between toxicants and environmental conditions, it is important to understand the influence of these conditions on the factors described above. The

underlying mechanisms for the observed interactions involve both physical and physiological processes.

B. Bioavailability

The studied environmental factors may influence the bioavailability of substances. For aquatic organisms, exposure to toxicants mainly occurs via the ambient water. As the solubility of compounds is temperature dependent, varying water temperature influences the bioavailability of toxicants. Food level may determine availability of chemicals as they may adsorb to particulate organic matter, such as algae. The route of exposure determines whether toxicity increases or decreases. Salinity is especially important for the bioavailability of metals. In freshwater, metals mainly exist in the dissociated and most available form, while at higher salinity metals are bound to ligands. Indeed, the few studies that used free metal ion concentration showed either no salinity effect or minimum toxicity at intermediate salinity (Lin and Dunson, 1993; Wildgust and Jones, 1998). Another interaction between metals and salinity is competition of metals with calcium and magnesium at uptake sites, resulting in decreased bioavailability at high salinity (McLusky et al., 1986). The amount of compound accumulated by organisms is also determined by the residence time of the chemical. Salinity may influence this process, as the degradation half-life of an organophosphate insecticide was higher in saline water relative to freshwater (Somasundaram and Coats, 1991; Walker, 1976, both in Ordelman et al., 1994).

C. Toxicokinetics

Temperature may alter physical factors such as partition coefficients or diffusion rates, in that way affecting toxicokinetics. Modifications in the partitioning of compounds over different body parts may result in higher accumulation of the toxicant at the target site, leading to lower effect concentrations (Van Wezel and Jonker, 1998). Increased diffusion rates may result into higher toxicant uptake rates. Physiological processes may

influence toxicokinetics as well. In ectotherms, an increase of ambient temperature leads to increased metabolic rate. High food or nutrient levels may also raise metabolism in organisms. This may result in higher toxicant uptake rates. Elimination and detoxication mechanisms may counteract the thermal effect on uptake rate, as the rates of these processes may increase as well. The regularly reported increase in toxicity when temperature rises or nutritional condition improves suggests that the temperature effect on uptake rates is of more importance than the effect on elimination and detoxication rates. On the contrary, other processes might be involved as well, and it is unclear to which extent increased uptake rates contribute to the elevated toxicity.

The effects of salinity on uptake rates are complicated because osmoregulation plays a major role. Wright (1995) discussed the influence of salinity on metal uptake in estuarine species. In most studies reviewed, a negative relationship between salinity and uptake rates was observed. Besides salinity effects on metal speciation, metals affected the ionic balance in crustaceans and fish. Gills seem to be the primary site for metal uptake and high concentrations of ion transporting chloride cells are present in the gills. This and an apparent competition between metals and calcium made Wright suggest a common route of uptake for metals and calcium. The degree to which metal and calcium uptake are coupled may differ between species as well as within species, because calcium needs vary between different developmental stages. Vonck (1999) also reported species-specific effects of salinity on chloride cell density. In addition, Rainbow (1995) reported that some crustaceans could vary metal uptake rates by changing their water permeability. Different ionic selectivity of the gills at low and high salinity, as well as lower apparent water permeability of crabs acclimated to low salinity were observed by Laporte et al. (1997). It was unclear how these factors influence metal uptake. Some studies (in Wright [1995]) mentioned urine as a metal excretory route, especially in freshwater animals where osmotic inflow of water is compensated by increased urine production. As literature is scarce, the effect of this factor on metal accumulation is unclear.

In contrast to metals, the influence of salinity on uptake of other chemicals was unclear. As

salinity does not affect speciation of these chemicals, this subscribes the idea that metal uptake rates are determined by speciation. At the isosmotic point, the uptake of compounds other than metals is likely to be decreased, as the inflow and drinking of water and ion exchange is reduced. Thomas and Rice (1986) described the influence of drinking rate on the accumulation of organic chemicals. They measured higher toluene and naphthalene accumulation in the salmonid dolly varden (*Salvelinus malma*) in saline water (30‰) compared with freshwater. This was associated with the increased drinking rate of fish in seawater, resulting in a shorter stomach emptying time and increased uptake of the chemicals from the gut. The elimination of the hydrocarbons by the gills and the urine was also lower in seawater.

D. Sensitivity of Organisms

Variations in environmental factors may also change toxic effect concentrations by altering the sensitivity of target sites, as was reported for temperature (Fisher and Wadleigh, 1985). On the other hand, many organisms are able to regulate the internal toxicant concentration, either by increased elimination, detoxication, or storage in specific body parts. Because these regulation mechanisms cost energy, less energy is available for other processes (Smit and Van Gestel, 1997). It is often assumed in bioenergetics that net assimilated energy is first allocated to maintenance, leaving less energy available for growth or reproduction (Calow, 1979; Koehn and Bayne, 1989; Kooijman, 1993; Kooijman and Metz, 1984; Nisbet and Gurney, 1989). Geyer et al. (1985, 1990, 1993a,b) observed less negative effects in animals with high fat content relative to animals with low fat content. This indeed indicates that well-fed organisms exposed to toxicants can pay the energy costs for coping with toxic effects more easily than those held at poor food conditions. Maternal food also proved to influence the sensitivity of juvenile daphnids to toxicants (Baird et al., 1989; Enserink et al., 1990, 1993). The maternal lipid reserves are higher in large, less sensitive juveniles (born from poor-fed mothers) compared with small, sensitive ones (born from well-fed mothers) (Tessier and Consolatti, 1989),

which may be associated with the available energy to withstand toxic stress. When organisms are exposed to narcotic chemicals for a short time period, adverse effects may also be lower in individuals with a higher fat content because a larger amount of the compound is stored in the fat compartment, leaving less available to reach the target site. Osmotic stress may also induce lower tolerance to chemicals because of increased energy costs for osmoregulation (i.e., maintenance costs).

E. Quantitative Relationships

Some of the interactions between chemicals and environmental factors that were described above were also analyzed quantitatively. The calculated coefficients suggest that effects of temperature on toxicity increase with exposure time. This can be understood because toxicant uptake rates are elevated at high temperatures due to increased metabolism. The difference in uptake rates at low and high temperatures results in an increase of toxic effects at the elevated temperature. This difference in toxicity becomes more apparent when the exposure time is extended. The influence of nutritional state on toxicity was determined by exposure time as well. Compared with organisms held at low food levels, well-fed organisms are in a better physiological condition, which makes them more resistant to toxic stress. The difference in toxicity increases when the organisms are exposed to the contaminant for a longer period of time.

The coefficients calculated for the influence of temperature, nutritional state, and salinity on toxicity give an indication of the magnitude of uncertainty factors applicable for laboratory to field extrapolations. Considering temperature, toxicity appears to be smaller than in the laboratory when extrapolating to ecosystems located in temperate regions, which would result in an uncertainty factor smaller than one. When extrapolating to systems nearby discharge points of cooling water, toxicity in the laboratory seems to be underestimated, resulting in an uncertainty factor greater than one. For nutritional state, toxicity under suboptimal conditions appears to be higher than expected from results of laboratory tests, hence the uncertainty factor should be greater

than one. An extreme salinity increase from freshwater to saline water results in an uncertainty factor smaller than one.

Chapman et al. (1998) state that uncertainty factors are often overprotective and they recommend a factor that does not exceed the value of 10 and may be much less. Considering the coefficients calculated here, this factor of 10 appears to be sufficient in most situations and it even seems that organisms will be overprotected. However, such an uncertainty factor is not high enough to protect organisms to thermal effects when extrapolating EC50 (48 h) values to ecosystems exposed to higher temperatures than in the laboratory, while it is just sufficient for the extrapolation of LC50 (96 h) values obtained at favorable food levels to food conditions in an oligotrophic lake. Most lakes in the Netherlands are eutrophic and the food conditions in those lakes will be more favorable. This does not imply that the nutrition-toxicant interaction is unimportant in eutrophicated waters, because food scarcity may even occur in eutrophicated waters, when populations have reached their carrying capacity (Buikema et al., 1980 [in Klüttgen et al., 1996]; Kooijman and Metz, 1984). Even if problems with toxicants are absent, they may become manifest when algal concentrations decrease due to declined nutrient inputs. This might be useful to consider before investing effort in reducing nutrient concentrations in surface waters. The factor of 10 seems to be high enough to translate results from standard tests conducted in freshwater to marine systems. However, because freshwater organisms do not occur in marine waters, this extrapolation is irrelevant. When extrapolating toxicity data to freshwaters with elevated salt concentrations, the toxicity change is probably much smaller. Further, as the interaction between salinity and toxicants is not clear, it remains uncertain what a sufficient value of the uncertainty factor will be.

As data were scarce, the coefficients only give an indication of the influence of the environmental factors on toxicity and the outcomes should not be overvalued. Table 1 shows that the choice for an uncertainty factor greater or smaller than one is determined by many factors. In this review only one factor was studied: the relation between toxicity of compounds and some environmental conditions.

F. Final Considerations and Research Needs

It is good to realize that if organisms are exposed to extremely stressful environmental conditions, this may result in such adverse effects that additional stress by chemicals may be overruled. Problems may arise when using quantitative endpoints such as growth or reproduction. Further, many studies reviewed used mortality as an endpoint. The lethal toxicant concentrations used in these experiments are often too high to be environmentally realistic. However, the observed interactions were generally the same for studies using lethal and sublethal endpoints, suggesting that the type of correlation between toxicity and environmental factors is independent of the toxicant concentration. In addition, the interaction between environmental factors and toxicants is not constant in natural systems, as environmental conditions, toxicant concentrations and the physiological state and metabolic needs of organisms may change in time. Nevertheless, the present review shows that environmental factors may modify the toxicity of chemicals to aquatic organisms. Organisms living near their environmental tolerance limits are more susceptible to additional stressors like toxicants and concentrations that are sublethal to organisms living at optimal environmental conditions may become lethal at sub-optimal conditions.

In order to obtain uncertainty factors that sufficiently protect natural systems without being overprotective, additional research is necessary. Because elevated temperatures accelerate both toxicokinetics and physiological processes, exposure time should be based on a physiological time scale instead of using fixed test durations prescribed by standard procedures (Donker et al., 1998). This means that exposure times should be adjusted for the different temperatures tested, especially when sublethal parameters such as growth and reproduction are considered. The influence of acclimating an organism to new experimental conditions before performing the toxicity test should also be considered. For instance, acclimation may decrease the effect of temperature on toxicity, as enzymes are slightly modified by temperature changes. This process takes time, depending on species and body size (Kooijman, 1993). Instead

of what was usually reported in the reviewed articles, studies performed to investigate the influence of salinity on metal toxicity should include the measurement of the free ion concentration. This would improve the ability to detect mechanisms in the case of changes in toxic effects. Both studies on individuals and populations are useful, as the first type of studies may provide an understanding of mechanisms involved in multiple stress interactions, while the second type is ecologically more relevant and will help to acquire more accurate uncertainty factors for the extrapolation of laboratory data to field situations. Environmental conditions may alter toxicokinetics, as described above. However, it is still unclear to which extent uptake, elimination, and detoxication rates are modified. Measurement of internal toxicant concentrations may help to obtain more insight into these processes and their influence on toxicity.

IX. CONCLUSIONS

The present review indicates that environmental factors may affect the toxicity of chemicals to aquatic organisms. Organisms living under environmental conditions that are near to their tolerance limits are often less resistant to additional stressors such as exposure to pollution. Generally, toxicity increased with increasing temperature and decreasing food or nutrient supply. Metal toxicity increased with decreasing salinity. To a great extent this was related to chemical speciation. On the other hand, toxicity of organophosphate insecticides increased at higher salinity, possibly due to differences in degradation rates. The influence of salinity on the toxicity of other chemicals was unclear. Both physical and physiological processes influencing factors like bioavailability, toxicokinetics and sensitivity of organisms should be considered when evaluating these interactions. As standard toxicity tests do not take variable and unfavorable experimental conditions into account, results of these tests are less realistic for toxicity in field situations.

Quantitative analysis of data reveals that an uncertainty factor for laboratory to field extrapolations should be smaller than one when extrapolating to ecosystems in temperate regions,

while it should be greater than one for systems receiving discharges of cooling water. When considering nutritional effects, the factor should be greater than one. On the other hand, the factor should be smaller than one when translating results of standard toxicity tests conducted in freshwater to waters with elevated salt concentrations. The toxicity under laboratory and relevant field conditions differed by a factor of 2.6 to 130 and 1.3 to 15 for the two temperature conditions

and 1.2 to 10 for nutritional state, dependent on the effect parameter considered. An extreme salinity change from freshwater to marine water decreased toxicity by a factor of 2.1, but as less extreme salinity changes are more relevant under field conditions, the change in toxicity is probably much smaller. Additional research is necessary to obtain uncertainty factors that will sufficiently protect ecosystems without being overly conservative.

ANNEX 1

Number of Experiments where Positive,¹ Negative,² No Correlations,³ or Optima⁴ Were Reported for Joint Effects of Toxicants and Temperature on Mortality

Toxicant	Positive correlation	Negative correlation	Optimum	No correlation
Metals				
Cadmium	15			2
Chromium	3			2
Copper	6			2
Lead	2			1
Magnesium	1			
Mercury	7	4	1	4
Nickel	3			2
Zinc	30		1	2
Pesticides				
Chlorinated hydrocarbons	18	7	1 ⁵	
Herbicides	2			
Organophosphates	1	1		
Natural toxicants				
Ammonia	2			2
Other toxicants				
Cyanide	4	1		1
Phenols	1	1		
Synthetic detergents	3	1		2

¹ Toxicity increased with increasing temperature.

² Toxicity increased with decreasing temperature.

³ No clear relationship between toxicity and temperature.

⁴ Maximum toxicity at intermediate temperatures.

⁵ Maximum toxicity at intermediate temperatures.

Summarized from Cairns et al. (1975) and McLusky et al. (1986).

ANNEX 2

Overview of Positive,¹ Negative,² and No Correlations³ (None) between Toxicity and Temperature

Toxicant	Species	Mortality	CTM	Reference
Metals				
Cadmium	<i>Lepomis cyanellus</i>		None	Carrier and Beitinger (1984)
Cadmium	<i>Macrobrachium rosenbergii</i>		Negative	Rosas and Ramirez (1993)
Chromium	<i>Macrobrachium rosenbergii</i>		Negative	Rosas and Ramirez (1993)
Copper	<i>Actinonaias pectorosa</i>	Positive		Jacobson <i>et al.</i> (1997)
Dichromate	<i>Daphnia magna</i>	Positive		Persone <i>et al.</i> (1989)
Dichromate	<i>Tisbe longicornis</i>	Positive		Larrain <i>et al.</i> (1998)
Nickel	<i>Oncorhynchus kisutch</i>		Negative	Becker and Wolford (1980)
Nickel	<i>Salmo gairdneri</i>		Negative	Becker and Wolford (1980)
Pesticides				
Aldicarb	<i>Aedes aegypti</i>	Positive		Song <i>et al.</i> (1997)
Aldicarb	<i>Daphnia magna</i>	Positive		Song <i>et al.</i> (1997)
Dimethoate	<i>Aedes aegypti</i>	None		Song <i>et al.</i> (1997)
Dimethoate	<i>Daphnia magna</i>	None		Song <i>et al.</i> (1997)
Endothal	<i>Notropis lutrensis</i>		None	Take <i>et al.</i> (1983)
Imidacloprid	<i>Aedes aegypti</i>	None		Song <i>et al.</i> (1997)
Imidacloprid	<i>Daphnia magna</i>	None		Song <i>et al.</i> (1997)
Lindane	<i>Chironomus riparius</i>	Positive		Fisher and Wadleigh (1985)
Tebufenozide	<i>Aedes aegypti</i>	Positive		Song <i>et al.</i> (1997)
Tebufenozide	<i>Daphnia magna</i>	Positive		Song <i>et al.</i> (1997)
Natural toxicants				
Ammonia	<i>Penaeus setiferus</i>		None	Alcaraz <i>et al.</i> (1997)
Ammonia/nitrite	<i>Penaeus setiferus</i>		Negative	Alcaraz <i>et al.</i> (1997)
Cyanobacteria	<i>Daphnia pulex</i> (clone 1)	None		Hietala <i>et al.</i> (1997)
Cyanobacteria	<i>Daphnia pulex</i> (clone 2)	None		Hietala <i>et al.</i> (1997)
Nitrite	<i>Penaeus setiferus</i>		Negative	Alcaraz <i>et al.</i> (1997)
Other toxicants				
Phenol	<i>Camptostoma anomalum</i>		Negative	Chagnon and Hlohowskyj (1989)
Laurylsulfate	<i>Daphnia magna</i>	Positive		Persone <i>et al.</i> (1989)
1,2,4-Trichlorobenzene	<i>Gasterosteus aculeatus</i>	Positive		Van Wezel and Jonker (1998)

¹ Toxicity increased with increasing temperature.

² Toxicity increased with decreasing temperature.

³ No clear relationships between toxicity and temperature.

Note: CTM = Critical Thermal Maximum.

ANNEX 3 Overview of Positive,¹ Negative,² and No Correlations³ (None) between Toxicity and Nutritional State

Effect parameter						
Toxicant	Species	Nutritional factor	Growth	Reproduction	Mortality	Ref.
Metals						
Cadmium	<i>Daphnia carinata</i>	Algae	Negative	Negative	Negative	Chandini (1988a)
Cadmium	<i>Daphnia carinata</i>	Algae	Positive (body length)	Positive (brood size)	Negative	Chandini (1989)
Cadmium	<i>Daphnia magna</i>	Algae	Negative (development juveniles)	Negative (age at maturity)	Negative	Klütgen and Ratte (1994)
Cadmium	<i>Daphnia magna</i>	Algae ⁴			Positive (acute test)	Enserink <i>et al.</i> (1990)
					None (chronic test)	
Cadmium	<i>Echinisca triserialis</i>	Algae		Negative	Negative	Chandini (1988a)
Cadmium	<i>Echinisca triserialis</i>	Algae			Negative	Chandini (1988b)
Chromium	<i>Daphnia magna</i>	Algae ⁴			None	Enserink <i>et al.</i> (1990 and 1993)
Copper	<i>Bosmina longirostris</i>	Algae			Negative	Koivisto <i>et al.</i> (1992)
Copper	<i>Chlamydomonas geitleri</i>	Nitrogen/phosphorus	None	Negative		Hall <i>et al.</i> (1989a)
Copper	<i>Chlorella vulgaris</i>	Nitrogen/phosphorus	None	Negative		Hall <i>et al.</i> (1989a,b)
Copper	<i>Chydorus sphaericus</i>	Algae			Negative	Koivisto <i>et al.</i> (1992)
Copper	<i>Daphnia galeata</i>	Algae			Negative	Koivisto <i>et al.</i> (1992)
Copper	<i>Daphnia magna</i>	Algae			Negative	Koivisto <i>et al.</i> (1992)
Copper	<i>Daphnia pulex</i>	Algae			Negative	Koivisto <i>et al.</i> (1992)
Copper	<i>Thalassiosira pseudonana</i>	Manganese	Negative			Sunda and Huntsman (1998)
Lead	<i>Daphnia magna</i>	Algae	Negative	Negative		Enserink <i>et al.</i> (1995)
Zinc	<i>Thalassiosira pseudonana</i>	Manganese	Negative			Sunda and Huntsman (1998)
Pesticides						
Endosulfan	<i>Daphnia carinata</i>	Algae	None	Positive (brood size)	None	Barry <i>et al.</i> (1995)
				Negative (age at maturity)		
				None (fecundity)		
Esfenvalerate	<i>Daphnia carinata</i>	Algae	None	None	Negative	Barry <i>et al.</i> (1995)

Natural toxicants					
Cyanobacteria	<i>Brachionu scalyCIFlorus (clone 1)</i>	Algae ⁵	None	Negative	Gilbert (1996)
Cyanobacteria	<i>Brachionus c alyCIFlorus (clone 2)</i>	Algae ⁵	None	None	Gilbert (1996)
Cyanobacteria	<i>Daphnia pulex</i>	Algae ⁵	Negative		Reinikainen <i>et al.</i> (1995)
Cyanobacteria	<i>Daphnia pulex</i>	Algae ⁵		Negative	Reinikainen <i>et al.</i> (1998)
Cyanobacteria	<i>Daphnia pulex</i>	Algae ⁵		Negative	Walls <i>et al.</i> (1997)
Cyanobacteria	<i>Synchaeta pectinata</i>	Algae ⁵	None	Negative	Gilbert (1996)
Kairomone	<i>Daphnia ambigua</i>	Algae	Negative	Negative	Hanazato (1991a)
Kairomone	<i>Daphnia pulex</i>	Algae	None	Positive	Walls <i>et al.</i> (1997)
Other toxicants					
3,4-dichloroaniline	<i>Ceriodaphnia quadrangula</i>	Algae	Positive (age at maturity) Negative (brood size)		Klüttgen <i>et al.</i> (1996)
3,4-dichloroaniline	<i>Daphnia magna</i>	Algae	Negative		Klüttgen <i>et al.</i> (1996)
3,4-dichloroaniline	<i>Daphnia magna</i>	Algae ⁴		Positive	Baird <i>et al.</i> (1989)

¹Toxicity increased with increasing food or nutrient level with respect to effect parameter.

²Toxicity increased with decreasing food or nutrient level with respect to effect parameter.

³No relationship between toxicity and food or nutrient level.

⁴Algae concentration at which mothers of test animals were cultured.

⁵Additional algae concentration added.

ANNEX 4

Number of Experiments Where Positive,¹ Negative,² No Correlations³ or Optima⁴ Were Reported for Joint Effects of Toxicants and Salinity on Mortality

Toxicant	Positive correlation	Negative correlation	Optimum	No correlation
Metals				
Arsenic				3
Cadmium	2	25	4	7
Chromium		6		
Copper	2	6	1	2
Lead		2		1
Mercury		10	4	2
Nickel		7		
Silver				1
Zinc	1	7	3	1
Pesticides				
Biocides	4	6	1	
Herbicides	1	1		
Organophosphates	11	1		5
Chlorinated hydrocarbons	5	7	1	4
Pyrethroids	1	1	2	
Carbamates		1		1
Other toxicants				
Petroleum hydrocarbons				1
Polycyclic aromatic hydrocarbons	2	1	1	1
Organic chemicals	2	6	2	1

¹Toxicity increased with increasing salinity.

²Toxicity increased with decreasing salinity.

³No relationship between toxicity and salinity.

⁴Minimum toxicity at intermediate salinity.

Summarized from Hall and Anderson (1995) and McLusky *et al.* (1986).

ANNEX 5

Overview of Positive,¹ Negative,² No Correlations³ (None) or Optima⁴ between Toxicity and Salinity

Toxicant	Species	Mortality	Ref.
Metals			
Cadmium	<i>Rivulus marmoratus</i>	Negative	Lin and Dunson (1993)
Cadmium (free ion)	<i>Rivulus marmoratus</i>	None	Lin and Dunson (1993)
Cadmium	<i>Lates calcarifer</i>	Negative	Shazili (1995)
Cadmium (free ion)	<i>Neomysis integer</i>	Optimum	Wildgust and Jones (1998)
Pesticides			
Aldicarb	<i>Aedes taeniorhynchus</i>	Positive	Song and Brown (1998)
Aldicarb	<i>Artemia</i> sp.	Positive	Song and Brown (1998)
Dimethoate	<i>Aedes taeniorhynchus</i>	Positive	Song and Brown (1998)
Dimethoate	<i>Artemia</i> sp.	Positive	Song and Brown (1998)
Imidacloprid	<i>Aedes taeniorhynchus</i>	Positive	Song and Brown (1998)
Imidacloprid	<i>Artemia</i> sp.	Positive	Song and Brown (1998)
Tebufenozide	<i>Aedes taeniorhynchus</i>	Positive	Song and Brown (1998)
Tebufenozide	<i>Artemia</i> sp.	Positive	Song and Brown (1998)
Other toxicants			
1,2,4-Trichlorobenzene	<i>Gasterosteus aculeatus</i>	None	Van Wezel and Jonker (1998)
Mixture of organic and inorganic chemicals	<i>Morone saxatilis</i>	Negative	Palawski <i>et al.</i> (1985)

¹Toxicity increased with increasing salinity.

²Toxicity increased with decreasing salinity.

³No relationship between toxicity and salinity.

⁴Minimum toxicity at intermediate salinity.

ANNEX 6 Quantitative Relations between Toxicity of Chemicals and Temperature, Nutritional State and Salinity or Combinations of These Factors

Factor	Species	Toxicant	Effect parameter	Exposure time	Number of data points for regression	Intercept (α_1 or α_2)	Coefficient for environmental factor ($\beta_1, \gamma_1, \gamma_2$ or γ_3)	Coefficient for toxicant (%)	R-square	Ref.
Temperature	<i>Daphnia magna</i>	Aldicarb	LC50	48 h	2	-42.7	$1.25 * 10^4$	N.S.		Song <i>et al.</i> (1997)
	<i>Actinonaias pectorosa</i>	Copper	LC50	24 h	4	-7.60	$2.75 * 10^3$	0.991		Jacobson <i>et al.</i> (1997)
	<i>Daphnia pulex</i>	Cyanobacteria	EC50	48 h	2	-32.8	$9.70 * 10^3$	N.S.		Hietala <i>et al.</i> (1997)
	<i>Daphnia pulex</i>	Cyanobacteria	EC50	48 h	2	-45.8	$1.35 * 10^4$	N.S.		Hietala <i>et al.</i> (1997)
	<i>Daphnia pulex</i>	Cyanobacteria	EC50	48 h	2	-61.7	$1.81 * 10^4$	N.S.		Hietala <i>et al.</i> (1997)
	<i>Daphnia pulex</i>	Cyanobacteria	EC50	48 h	2	-53.3	$1.56 * 10^4$	N.S.		Hietala <i>et al.</i> (1997)
	<i>Daphnia pulex</i>	Cyanobacteria	EC50	48 h	2	-45.5	$1.34 * 10^4$	N.S.		Hietala <i>et al.</i> (1997)
	<i>Daphnia pulex</i>	Cyanobacteria	EC50	48 h	2	-38.4	$1.12 * 10^4$	N.S.		Hietala <i>et al.</i> (1997)
	<i>Daphnia pulex</i>	Cyanobacteria	EC50	48 h	2	-22.4	$6.31 * 10^3$	N.S.		Hietala <i>et al.</i> (1997)
	<i>Daphnia pulex</i>	Cyanobacteria	EC50	48 h	2	-14.8	$4.07 * 10^3$	N.S.		Hietala <i>et al.</i> (1997)
	<i>Daphnia pulex</i>	Cyanobacteria	EC50	48 h	2	-8.62	$2.20 * 10^3$	N.S.		Hietala <i>et al.</i> (1997)
	<i>Daphnia magna</i>	Dichromate	EC50	24 h	4	-16.8	$4.63 * 10^3$	0.963		Persoone <i>et al.</i> (1989)
	<i>Daphnia magna</i>	Dichromate	EC50	24 h	4	-7.55	$2.24 * 10^3$	0.968		Persoone <i>et al.</i> (1989)
	<i>Daphnia magna</i>	Dichromate	EC50	24 h	4	-10.3	$3.11 * 10^3$	0.890		Persoone <i>et al.</i> (1989)
	<i>Daphnia magna</i>	Dichromate	EC50	24 h	4	-13.2	$3.98 * 10^3$	0.956		Persoone <i>et al.</i> (1989)
	<i>Tisbe longicornis</i>	Dichromate	LC50	48 h	3	-14.3	$4.47 * 10^3$	0.936		Larain <i>et al.</i> (1998)
	<i>Tisbe longicornis</i>	Dichromate	LC50	48 h	3	-21.3	$6.51 * 10^3$	0.943		Larain <i>et al.</i> (1998)
	<i>Tisbe longicornis</i>	Dichromate	LC50	48 h	3	-15.9	$4.96 * 10^3$	0.975		Larain <i>et al.</i> (1998)
	<i>Tisbe longicornis</i>	Dichromate	LC50	48 h	3	-17.0	$5.28 * 10^3$	0.969		Larain <i>et al.</i> (1998)
	<i>Tisbe longicornis</i>	Dichromate	LC50	48 h	3	-23.7	$7.20 * 10^3$	0.972		Larain <i>et al.</i> (1998)
	<i>Tisbe longicornis</i>	Dichromate	LC50	48 h	3	-18.6	$5.72 * 10^3$	0.962		Larain <i>et al.</i> (1998)
	<i>Daphnia magna</i>	Laurylsulfate	EC50	24 h	4	-14.9	$4.71 * 10^3$	0.951		Persoone <i>et al.</i> (1989)
	<i>Daphnia magna</i>	Laurylsulfate	EC50	24 h	4	-8.04	$2.80 * 10^3$	0.970		Persoone <i>et al.</i> (1989)
	<i>Daphnia magna</i>	Laurylsulfate	EC50	24 h	4	-11.4	$3.76 * 10^3$	0.921		Persoone <i>et al.</i> (1989)
	<i>Daphnia magna</i>	Laurylsulfate	EC50	24 h	4	-9.43	$3.21 * 10^3$	0.821		Persoone <i>et al.</i> (1989)
	<i>Chironomus riparius</i>	Lindane	LC50	24 h	3	-3.56	$1.33 * 10^3$	0.744		Fisher and Wadleigh (1985)
	<i>Gasterosteus aculeatus</i>	1,2,4-Tetrachlorobenzene (wet weight)	LBB		3	-4.26	$1.40 * 10^3$	0.945		Van Wezel and Jonker (1998)

Nutritional state	<i>Daphnia magna</i>	Cadmium	Body length	Chronic	16	-0.061	-0.0459	-0.0536	0.781	Klüttgen and Ratte (1994)
	<i>Daphnia magna</i>	Cadmium	Brood size	Chronic	16	0.581	-0.402	-0.442	0.865	Klüttgen and Ratte (1994)
	<i>Daphnia carinata</i>	Cadmium	LC50	48 h	2	2.66	0.253		N.S.	Chandini (1988a)
	<i>Daphnia carinata</i>	Cadmium	LC50	96 h	3	2.68	0.671		1.00	Chandini (1988a)
	<i>Echinisca triserialis</i>	Cadmium	LC50	48 h	3	2.65	0.131		0.919	Chandini (1988a)
	<i>Echinisca triserialis</i>	Cadmium	LC50	96 h	3	2.39	0.719		0.662	Chandini (1988a)
	<i>Echinisca triserialis</i>	Cadmium	Life span	Chronic	12	0.370	0.676	-0.503	0.859	Chandini (1988b)
	<i>Echinisca triserialis</i>	Cadmium	Longevity	Chronic	12	0.0937	0.182	-0.188	0.878	Chandini (1988b)
	<i>Echinisca triserialis</i>	Cadmium	Adult size	Chronic	12	0.0333	0.0740	-0.0788	0.813	Chandini (1988b)
	<i>Echinisca triserialis</i>	Cadmium	Size at maturity	Chronic	12	0.0435	0.0642	-0.0078	0.540	Chandini (1988b)
	<i>Echinisca triserialis</i>	Cadmium	Neonate size	Chronic	12	-0.00530	0.0164	-0.0275	0.280	Chandini (1988b)
	<i>Echinisca triserialis</i>	Cadmium	Fecundity	Chronic	12	0.800	0.962	-1.26	0.762	Chandini (1988b)
	<i>Daphnia carinata</i>	Cadmium	Life span	Chronic	12	0.935	0.550	-0.474	0.817	Chandini (1989)
	<i>Daphnia carinata</i>	Cadmium	Longevity	Chronic	12	0.949	0.360	-0.509	0.834	Chandini (1989)
	<i>Daphnia carinata</i>	Cadmium	Size at maturity	Chronic	6	1.33	0.188	-0.637	0.843	Chandini (1989)
	<i>Daphnia carinata</i>	Cadmium	Fecundity	Chronic	12	8.15	2.36	-4.14	0.724	Chandini (1989)
	<i>Daphnia pulex</i>	Cyanobacteria	Survival time	Chronic	6	1.34	0.234	-0.334	0.874	Reinikainen et al. (1998)
	<i>Daphnia pulex</i>	Cyanobacteria	Egg mortality	Chronic	6	-0.793	-0.238	0.363	0.685	Reinikainen et al. (1995)
	<i>Brachionus calyciflorus</i>	Cyanobacteria	Life span	Chronic	3	-0.179	0.455		0.807	Gilbert (1996)
	<i>Synchaeta pectinata</i>	Cyanobacteria	Life span	Chronic	3	0.00833	0.291		0.527	Gilbert (1996)
	<i>Synchaeta pectinata</i>	Cyanobacteria	Fecundity	Chronic	3	0.0162	0.744		0.768	Gilbert (1996)
	<i>Synchaeta pectinata</i>	Cyanobacteria	Life span	Chronic	3	-0.144	0.136		0.318	Gilbert (1996)
	<i>Synchaeta pectinata</i>	Cyanobacteria	Fecundity	Chronic	2	-0.220	1.06		N.S.	Gilbert (1996)
	<i>Daphnia carinata</i>	Endosulfan	First brood size	Chronic	5	-0.188	-0.162		0.664	Barry et al. (1995)
	<i>Daphnia carinata</i>	Endosulfan	Age at maturity	Chronic	5	-0.0120	-0.0361		0.452	Barry et al. (1995)
	<i>Daphnia pulex</i>	Kairomone	Age at maturity	Chronic	2	0.0325	0.0644		N.S.	Walls et al. (1997)
	<i>Daphnia pulex</i>	Kairomone	Fecundity	Chronic	2	-0.123	-0.200		N.S.	Walls et al. (1997)
	<i>Daphnia magna</i>	Kairomone	Age at maturity	Chronic	9	0.0569	-0.0936		0.501	Klüttgen et al. (1996)
	<i>Daphnia magna</i>	Kairomone	Fecundity	Chronic	6	0.720	0.443	-0.891	0.662	Klüttgen et al. (1996)

ANNEX 6 (continued)

Salinity	<i>Ceriodaphnia quadrangula</i>	Kairomone	age at maturity	chronic	9	-0.0496	-0.0555	-0.0298	0.477	Klüttgen <i>et al.</i> (1996)
	<i>Ceriodaphnia quadrangula</i>	Kairomone	fecundity	chronic	6	0.291	-0.0320	-1.12	0.546	Klüttgen <i>et al.</i> (1996)
Salinity	<i>Rivulus marmoratus</i>	Cadmium	LC50	96 h	2	1.13	0.166		N.S.	Lin and Dunson (1993)
	<i>Lates calcarifer</i>	Cadmium	LC50	96 h	3	-0.551	1.31		0.926	Shazili (1995)
	<i>Lates calcarifer</i>	Cadmium	LC50	12 d	3	-1.24			0.940	Shazili (1995)
	<i>Lates calcarifer</i>	Cadmium	LC50	21 d	2	-1.55	1.66		N.S.	Shazili (1995)
	<i>Morone saxatilis</i>	Cadmium	LC50	96 h	2	1.88	0.318		N.S.	Palawski <i>et al.</i> (1985)
	<i>Neomysis integer</i>	Cadmium (free ion)	LC50	96 h	3	0.994	-0.0339		0.000150	Wildgust and Jones (1998)
	<i>Morone saxatilis</i>	Carbaryl	LC50	96 h	2	3.36	0.120		N.S.	Palawski <i>et al.</i> (1985)
	<i>Morone saxatilis</i>	Chromium	LC50	96 h	2	4.76	0.0791		N.S.	Palawski <i>et al.</i> (1985)
	<i>Morone saxatilis</i>	Copper	LC50	96 h	2	2.28	0.0697		N.S.	Palawski <i>et al.</i> (1985)
	<i>Morone saxatilis</i>	Malathion	LC50	96 h	2	1.81	0.106		N.S.	Palawski <i>et al.</i> (1985)
	<i>Morone saxatilis</i>	Nickel	LC50	96 h	2	4.32	0.183		N.S.	Palawski <i>et al.</i> (1985)
	<i>Morone saxatilis</i>	Selenium	LC50	96 h	2	3.19	0.0170		N.S.	Palawski <i>et al.</i> (1985)
	<i>Morone saxatilis</i>	Toxaphene	LC50	96 h	2	0.881	0.0371		N.S.	Palawski <i>et al.</i> (1985)
	<i>Morone saxatilis</i>	Zinc	LC50	96 h	2	2.63	0.139		N.S.	Palawski <i>et al.</i> (1985)
Nutritional state and salinity	<i>Thalassiosira pseudonana</i>	Chromium	EC50	Not reported	10	0.314	0.935 (nutrition) 0.982 (salinity)		0.987	Riedel (1984)
	<i>Thalassiosira pseudonana</i>	Chromium	EC50	Not reported	10	0.171	0.973 (nutrition) 1.08 (salinity)		0.966	Riedel (1984)

Temperature	<i>Praunus flexuosus</i>	Chromium	LC50	48 h	12	0.568	-0.262	(temperature)	0.892	McLusky and Hagerman (1987)
salinity	<i>Praunus flexuosus</i>	Chromium	LC50	96 h	12	0.330	0.824 (salinity)	(salinity)	0.857	McLusky and Hagerman (1987)
	<i>Praunus flexuosus</i>	Chromium	LC50	192 h	10	0.661	-0.357 (temperature)	(temperature)	0.562	McLusky and Hagerman (1987)
	<i>Artemia salina</i>	Dichromate	LC50	24 h	20	5.12	0.901 (salinity)	(salinity)	0.841	Persoone et al. (1989)
	<i>Brachionus plicatilis</i>	Dichromate	LC50	24 h	16	2.23	-0.322 (temperature)	(temperature)	0.796	Persoone et al. (1989)
	<i>Artemia salina</i>	Laurylsulfate	LC50	24 h	20	3.95	-0.422 (salinity)	(salinity)	0.892	Persoone et al. (1989)
	<i>Brachionus plicatilis</i>	Laurylsulfate	LC50	24 h	16	1.84	-2.60 (temperature)	(temperature)	0.827	Persoone et al. (1989)
							-0.0104 (salinity)	(salinity)		
							-0.250 (temperature)	(temperature)		
							0.447 (salinity)	(salinity)		
							-1.26 (temperature)	(temperature)		
							-0.582 (salinity)	(salinity)		
							-0.226 (temperature)	(temperature)		
							-0.234 (salinity)	(salinity)		

Data were fitted to equation (1) and (2) (see text). N.S. = not significant, as only two data points were available.

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